

NATO STANDARD

ALP - 16

EXPLOSIVES SAFETY AND MUNITIONS RISK MANAGEMENT (ESMRM) IN NATO PLANNING, TRAINING, AND OPERATIONS

**Edition A Version 1
APRIL 2015**



NORTH ATLANTIC TREATY ORGANIZATION

ALLIED LOGISTICS PUBLICATION

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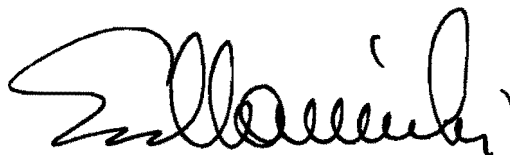
NORTH ATLANTIC TREATY ORGANIZATION (NATO)

NATO STANDARDIZATION OFFICE (NSO)

NATO LETTER OF PROMULGATION

30 April 2014

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Edvardas MAŽEIKIS
Major General, LTUAF
Director, NATO Standardization Office

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CHAPTER 1 NATO EXPLOSIVES SAFETY AND MUNITIONS RISK MANAGEMENT (ESMRM)

1.1. PURPOSE

The purpose of Allied logistic publication (ALP) - 16 is to establish requirements for the NATO ESMRM process and identify roles and responsibilities applicable to the NATO Operational Planning Process (OPP), the Operational Stages, and the Consumer Logistics Process across the full range of NATO military operations to include munitions-related contracted support.

1.2. ESMRM DEFINITION AND POLICY

1.2.1. ESMRM. ESMRM is a systematic approach that integrates risk assessment into operations planning, military training exercises, and contingency operations with the goal of identifying potential consequences associated with munitions operations, risk reduction alternatives, and risk decision criteria for key decision makers. ESMRM permits NATO commanders to deliver operational effect at tolerable levels of risk, and must be done on a recurring basis to be effective.

1.2.2. NATO Requirements. ESMRM and its supporting assessment and risk decision processes are based on the NATO operational planning process, explosives safety, and logistics requirements. Figure 1-1 illustrates the applicable NATO policies, doctrine and technical standards and guidelines.

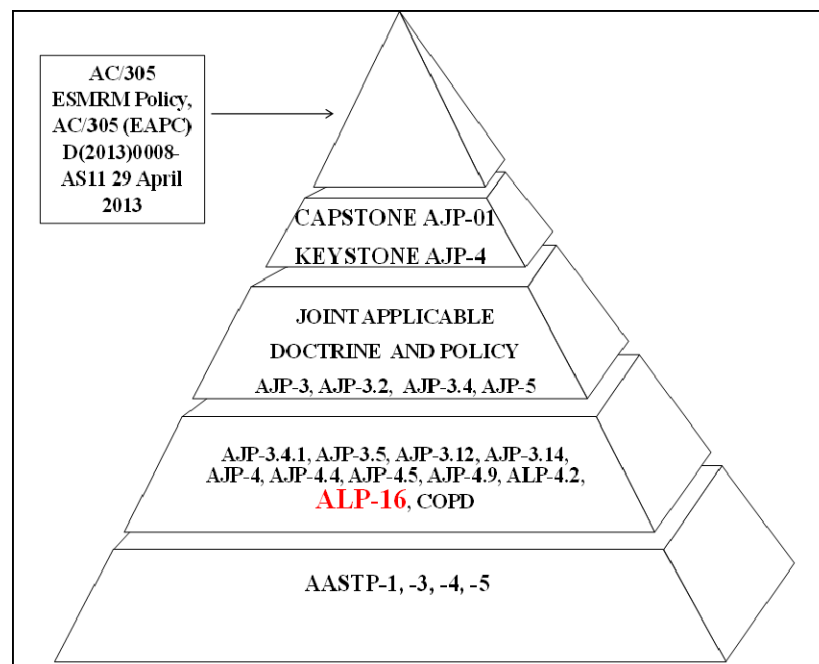


Figure 1-1. NATO ESMRM Authority, Doctrine, Policy, and Technical Requirements Hierarchy*

1.2.3. Policy

- a. NATO commanders and NATO nations participating in NATO planning, exercises, and operations shall comply with NATO explosives safety requirements¹ whenever possible.
- b. Nations that provide, transport, store, or otherwise handle munitions for NATO training and contingency operations have a responsibility to provide serviceable munitions that are well maintained and free from known defects that could lead to unintentional functioning.
- c. Munitions and munitions-related risk decisions by appropriate senior decision makers are both NATO commander and national responsibilities.
- d. When NATO is leading a coalition of partners, NATO shall identify, assess through the assessment process contained herein, and make risk decisions when the NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met.
- e. NATO must coordinate munitions risk assessments in advance of training and operations and communicate ESMRM risk decisions with host nations when transporting, storing, handling, and training with military munitions.
- f. NATO will coordinate, in advance of training and operations ESMRM risk assessments with the appropriate national authority(ies) responsible for accepting munitions or munitions-related risks when NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met.
- g. For the full range of NATO military operations involving one NATO nation, and when NATO explosives safety requirements can be met, that nation will record that the munitions operations meet the requirements. Subsequently, that nation will forward the document through the national chain of command to the appropriate level NATO headquarters (HQ).

1.2.4. ESMRM Relationship to Other NATO Functional Areas. ESMRM requirements shall be integrated throughout NATO doctrine and issuances as appropriate, since ESMRM requirements overlap and impact other functional areas. Examples of functional areas that have an ESMRM aspect include:

- a. Military engineers

¹ NATO explosives safety requirements are contained in Allied Ammunition Storage and Transport Publication (AASTP)-1, "Manual of NATO Safety Principles for Storage of Military Ammunition and Explosives," AASTP-3, "Manual of NATO Safety Principles for the Hazard Classification of Military Ammunition and Explosives," AASTP-4, "Manual on the NATO Principles for the Application of Risk Analysis to the Storage and Transportation of Military Ammunition and Explosives," and AASTP-5, "NATO Guidelines for the Storage, Maintenance and Transport of Ammunition on Deployed Missions or Operations."

- b. Base support operations
- c. Real property procurements and management
- d. Force protection
- e. Security

1.3. ESMRM SCOPE, APPLICABILITY, AND AUTHORITY

1.3.1. **Scope.** ALP-16 shall be used during the NATO OPP and throughout the full range of military operations, when munitions are involved, and establishes ESMRM requirements in the NATO:

- a. OPP
- b. Operational Stages
- c. Consumer Logistics Process

1.3.2. **Applicability.** ALP-16 is applicable to the full range of NATO military operations, both Article 5 crisis response operations and non-Article 5 crisis response operations (NA5CRO), as well as in exercises once the cooperation protocols have been agreed upon by the North Atlantic Council (NAC). The requirements of ALP-16 also apply to non-NATO nations participating in NATO-led operations and exercises.

- a. Full Range of NATO Military Operations. For the purposes of ALP-16, the full range of military operations includes NATO military missions and operations supporting Article 5 crisis response operations, NA5CRO, and other NATO support operations from planning through the disposal of supplies, to include munitions.
- b. ESMRM Applicability During Munitions Use. The requirements contained herein are not applicable when munitions are in use by forces (e.g., checkpoints or elements of a maneuvering force, engaged with the enemy or conducting movement to contact or support operations). In such cases, munitions and their related risks shall be addressed and managed by the appropriate NATO operational commander, consistent with mission requirements.
- c. Munitions-related Processes and Activities Performed Under Contract. Requirements of ALP-16 apply to other than military personnel (e.g. contract and host nation support) performing or supporting munitions-related functions on behalf of NATO nations in the area of operations (AOO).

1.4. ESMRM IN THE NATO OPP, OPERATIONAL STAGES, AND LOGISTICS

1.4.1. **ESMRM in the NATO OPP.** Just as logistics planning must be fully integrated into the OPP, the ESMRM process is critical to NATO mission success whenever munitions are involved. Chapter 3 provides details for integrating ESMRM in the NATO OPP and as part of logistics planning when munitions are involved.

1.4.2. **NATO Operational Stages.** A NATO operation usually consists of a number of stages as shown in Figure 1-2. These stages, which may overlap depending on the situation and mission, are Planning, Preparation for Deployment, Deployment, Employment, Transition and Redeployment, and Disestablishment. ALP-16 connects the ESMRM process to the NATO Operational Stages (see Table 1-1).

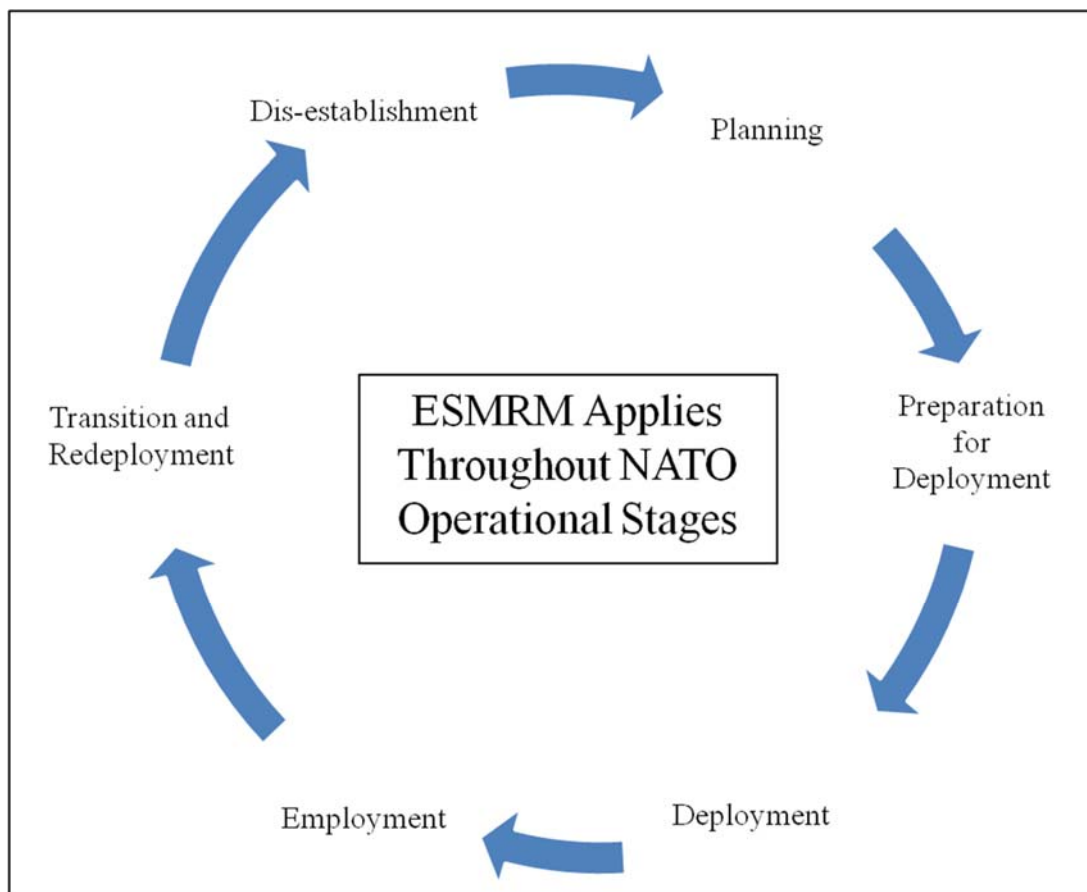


Figure 1-2. NATO Operational Stages

**Table 1-1. ESMRM Links to the NATO OPP,*
Operational Stages, and Consumer Logistics Process**

NATO Operational Planning Process	NATO Operational Stage	NATO Consumer Logistics Process	ESMRM Risk Assessment Type	ALP-16 Reference
Strategic Operations Planning	-	-	Planning	Chapter 3 and Chapter 7 Paragraph 7.3
Operational- Level Planning	-	-	Planning	Chapter 3 and Chapter 7 Paragraph 7.3
Tactical Operations Planning	-	-	Planning	Chapter 3 and Chapter 7 Paragraph 7.3
-	Deployment	Reception	Reception, Staging, Onward Movement and Integration (RSOI)	Chapter 4 Paragraph 4.3
-	Employment, Redeployment, and Disestablishment	Storage	Storage	Chapter 4 Paragraph 4.4
-	Deployment, Employment, Redeployment, and Disestablishment	Transportation	Transportation	Chapter 4 Paragraph 4.5
-	Employment	Distribution	Distribution	Chapter 4 Paragraph 4.6
-	Employment	Maintenance	Maintenance and Handling	Chapter 4 Paragraph 4.7
-	Redeployment and Disestablishment	Retrograde	Retrograde and Removal	Chapter 4 Paragraph 4.8
-	Redeployment and Disestablishment	Disposal	Disposal and Demilitarization	Chapter 4 Paragraph 4.9

*AJP-5, Figure 3.6, page 3-63.

1.4.3. NATO ESMRM Chain of Command. For the purposes of ALP-16, chain-of-command responsibilities and decision making associated with ESMRM, specified in Chapter 2, are integrated into existing NATO operational chain-of-command levels.

1.4.4. NATO Logistics. NATO Logistics encompasses the broad range of sustainment activity. It includes logistics management in peacetime as well as the logistics support of operation. A strong link exists between ESMRM and logistics since munitions and munitions-related operations are typically part of the logistics process. ALP-16 connects ESMRM to the NATO Consumer Logistics Process (see Table 1-1). Chapter 4 identifies ESMRM requirements related to the NATO Consumer Logistics Process.

1.5. ESMRM PROCESS

1.5.1. ESMRM Categories. The ESMRM process consists of two categories of ESMRM risk assessments: Planning and Operations/Execution. Combined, these ESMRM risk assessments and their risk decision integrate munitions-specific risk requirements throughout the full range of NATO military operations. As illustrated in Figure 1-3, the ESMRM process includes:

- a. Explosives Safety Criteria Assessment. The Explosives Safety Criteria Assessment is a determination of the ability to meet requirements of AASTP-1 or AASTP-5, as applicable.²
- b. ESMRM Risk Assessment. The ESMRM risk assessment is conducted when applicable NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met. The two broad types of ESMRM risk assessments include:
 - (1) Planning. Chapter 3 provides ESMRM risk assessment requirements for the NATO OPP. The three types of ESMRM risk assessments for the NATO OPP include:
 - (a) Strategic
 - (b) Operational
 - (c) Tactical
 - (2) Operations/Execution. Chapter 4 describes the ESMRM risk assessment types that link the NATO Operational Stages to the NATO Consumer Logistics Process phases. Chapter 7 provides the specific requirements for the seven types of ESMRM risk assessments, which include:
 - (a) RSOI
 - (b) Storage
 - (c) Transportation

² Where nations have their own explosives safety regulations, it is their responsibility to assess the equivalency of those regulations to NATO explosives safety requirements (see subparagraph 1.2.3.a and footnote 1).

- (d) Distribution
 - (e) Maintenance and Handling
 - (f) Retrograde and Removal
 - (g) Disposal and Demilitarization
- c. Risk Decision. All ESMRM risk assessments require a risk decision by the appropriate NATO and involved national operational chain-of-command level. Chapter 2 identifies risk assessment reviews, coordination requirements, and appropriate risk decision makers based on the risk level associated with the risk decision to be made.
- d. Risk Decision Implementation. Approved mitigation and conditions associated with the risk decision shall be implemented and managed. Chapter 2 identifies roles and responsibilities for ESMRM-related personnel who would implement and monitor required mitigation and conditions associated with the risk decision.
- e. Oversight. The risk decision, and any associated mitigation and conditions, shall be managed to ensure they continue to be implemented and remain current, correct, and visible at all levels, to include appropriate decision makers. Chapter 2 details chain-of-command and ESMRM personnel risk decision oversight responsibilities, periodic risk assessment reviews, and risk decision awareness of new NATO commanders.



Figure 1-3. NATO ESMRM Process in Relation to OPP, Exercise Planning, and Operations

1.5.2. ESMRM Assessment Development. ESMRM risk assessments shall be conducted by qualified staff. When not available, the NATO commander may request external assessment support through NATO and national chains of command. Request for NATO support should be made to NATO AC/305 (Logistics Committee (LC)), which in turn can request support from its ESMRM Panel, and NATO AC/326, Conference of NATO Armaments Directors' Ammunition Safety Group.

1.6. ESMRM ROLES AND RESPONSIBILITIES

1.6.1. An effective ESMRM program requires participation of key personnel throughout the NATO and national chains of command. For the purposes of ALP-16, a new ESMRM function, Explosives Safety Board (ESB), and the following three new ESMRM personnel roles are established:

- a. ESB chair
- b. Lead explosives safety officer (Lead ESO)
- c. Base operations support integrator (BOSI)

1.6.2. Chapter 2 defines the following ESMRM roles and responsibilities:

- a. Supreme Headquarters Allied Powers Europe (SHAPE)
- b. NATO operational commander
- c. National operational commander
- d. NATO base commander
- e. NATO base safety officer
- f. ESB
- g. ESB chair
- h. Lead ESO
- i. BOSI³
- j. National ESO

³ Base Operations Support Integrator is a function applicable to a NATO base having two or more nations, conducted on behalf of the NATO base commander to provide oversight and coordination of base functions and real estate and real property-related activities on that base. This function plays an important role in the NATO ESMRM process detailed in Chapter 2.

CHAPTER 2 EXPLOSIVES SAFETY AND MUNITIONS RISK MANAGEMENT (ESMRM) COMMAND AND CONTROL

2.1. PURPOSE

The purpose of this chapter is to identify the NATO operational chain of command ESMRM roles, responsibilities, and requirements.

2.2. ROLE OF ESMRM IN MILITARY PLANNING, EXERCISES, AND OPERATIONS ACROSS THE FULL RANGE OF NATO MILITARY OPERATIONS

2.2.1. The ESMRM process identifies munitions and munitions-related risks during NATO planning, training, and across the full range of military operations to the NATO operational chain of command (Figure 2-1).

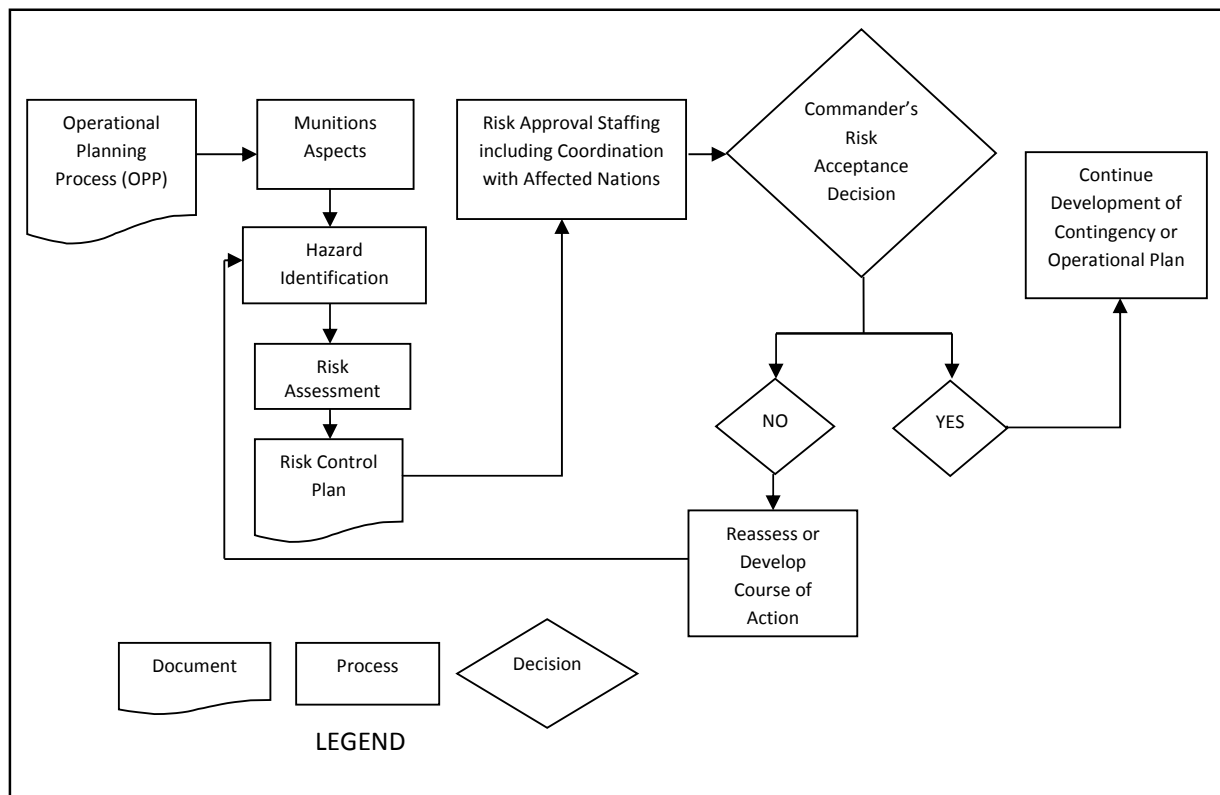


Figure 2-1. ESMRM Process

2.2.2. NATO explosives safety requirements¹, when properly applied, provide a tolerable level of risk to NATO personnel, assets, and facilities from NATO munitions and munitions-related processes.

2.2.3. The ESMRM process, when accomplished as part of the planning process and performed consistently in training and during execution, is an enabler that directly contributes to mission success. The process provides all levels of leadership with greater visibility of risks and potential consequences associated with munitions and munitions-related processes across the full range of NATO military operations.

2.2.4. The ESMRM process also provides information necessary to make an informed risk decision for the potential consequences when the risk cannot be mitigated to a level equivalent to those in AASTP-1 or AASTP-5 or eliminated.

2.2.5. Lessons learned shall be provided to the Joint Analysis Lessons Learned Centre (<https://nllp.jallc.nato.int/Pages/default.aspx>) to improve existing ESMRM policy and requirements.

2.2.6. Figure 2-2 illustrates the NATO ESMRM chain of command. Details are provided in paragraphs 2.3 through 2.7 below.

2.3. SUPREME HEADQUARTERS ALLIED POWERS EUROPE (SHAPE) ROLES AND RESPONSIBILITIES

2.3.1. NATO commanders shall integrate ESMRM into planning, training, and across the full range of NATO military operations. Integrating ESMRM early in the planning process enhances mission effectiveness while minimizing exposure to the potential consequences of munitions from accidental initiation or enemy fire.

2.3.2. SHAPE shall:

- a. Establish ESMRM positions and appoint qualified personnel to fill these positions.
- b. Direct the application of NATO explosives safety requirements.
- c. Conduct or coordinate the development of ESMRM risk assessments as part of strategic planning when NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met.
- d. Require recurring evaluations (between 12-36 months) of existing NATO ESMRM risk assessments and their risk decisions. (Chapter 7, paragraph 7.4 provides details.)

¹ NATO explosives safety requirements are contained in Allied Ammunition Storage and Transport Publication (AASTP)-1, "Manual of NATO Safety Principles for Storage of Military Ammunition and Explosives," AASTP-3, "Manual of NATO Safety Principles for the Hazard Classification of Military Ammunition and Explosives," AASTP-4, "Manual on the NATO Principles for the Application of Risk Analysis to the Storage and Transportation of Military Ammunition and Explosives," and AASTP-5, "NATO Guidelines for the Storage, Maintenance and Transport of Ammunition on Deployed Missions or Operations."

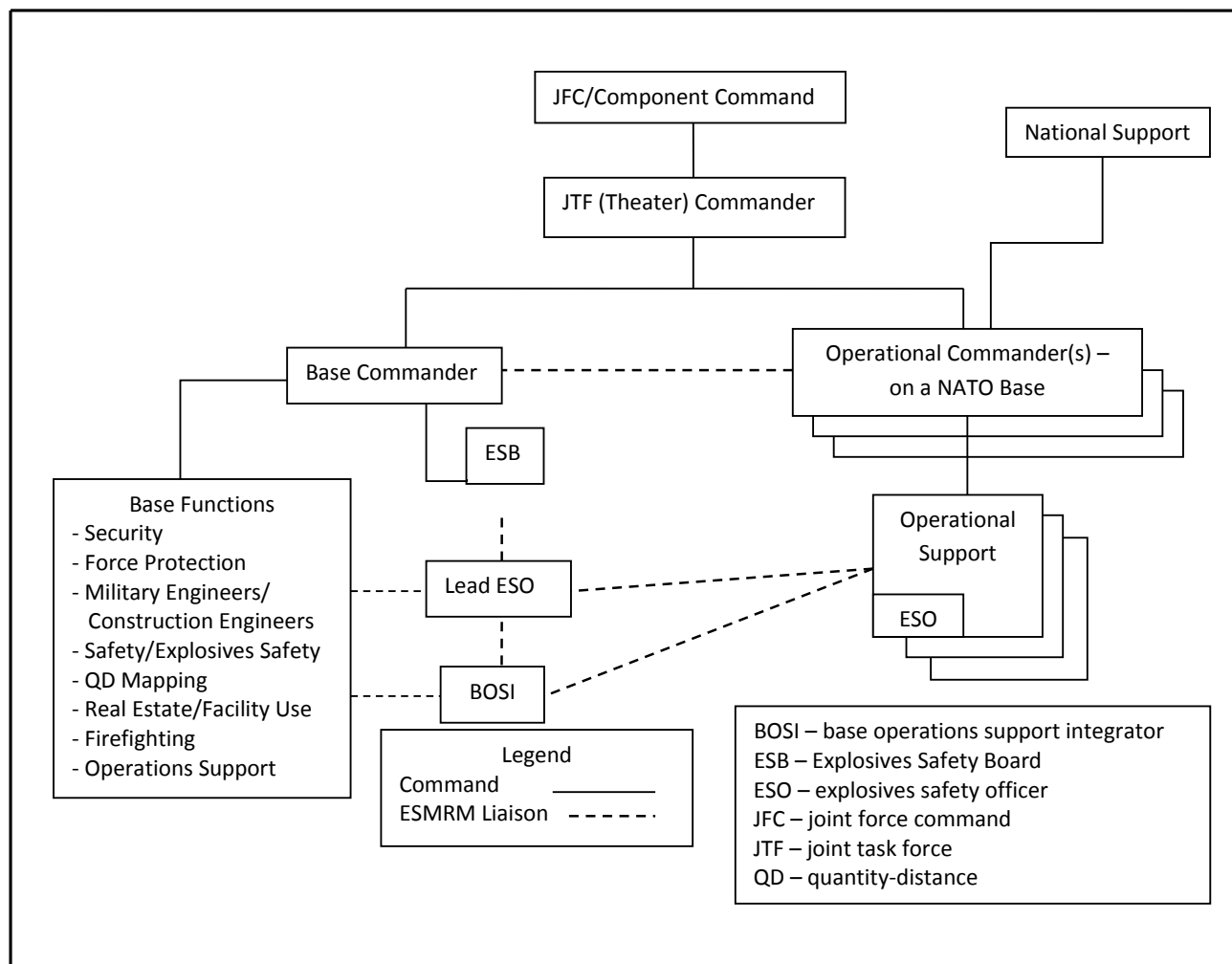


Figure 2-2. ESMRM Chain of Command

2.4. JFC AND COMPONENT COMMAND ROLES AND RESPONSIBILITIES

2.4.1. NATO commanders shall integrate ESMRM across the full range of NATO military operations to include planning. Integrating ESMRM early in the planning process enhances mission effectiveness while minimizing exposure to the potential consequences of munitions from accidental initiation or enemy fire.

2.4.2. All risk decisions involving munitions or munitions-related processes that do not meet NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, shall be based on ESMRM risk assessments as described in Chapters 4 and 7.

2.4.3. NATO JFC or component commanders are responsible for determining who will make the risk decision within their chains of command as specified in Chapter 2. If JFC or component commanders do not delegate the ESMRM risk decision authority, the JFC or component commander is responsible for making the risk decision.

2.4.4. NATO JFC or component commanders shall maintain awareness of munitions-related risks by periodically reviewing all risk decisions forwarded by JTF commanders.

2.4.5. The following apply to risk decision making, as shown in Chapter 7, Table 7-1:

- a. For risk assessments with high and medium risk levels within operational plans, a general or flag officer within the NATO chain of command is responsible for making the risk decision.
- b. For risk assessments with high and medium risk levels at an MN base requiring a risk decision at a higher level within the chain of command, subordinate NATO commanders shall endorse and forward to the appropriate NATO commander with risk decision authority.
- c. Risk assessments with a low risk level do not require approval delegation and can be approved by subordinate commanders.
- d. Risk decisions affecting other nations' personnel, assets, or missions, as well as any unrelated personnel to the munitions operation, shall always be made by a general or flag officer within the NATO operational chain of command.

2.4.6. For the purposes of Allied Logistic Publication (ALP)-16, general staff (e.g., G/J-3, G/J-4, etc.) are considered part of the NATO operational chain of command.

2.4.7. NATO JFC or component commanders shall:

- a. Make munitions-related risk decisions when delegated risk-decision authority following the process identified in subparagraph 2.4.5 and Chapter 7.
- b. Appoint ESMRM focal points (i.e., qualified explosives safety-related subject matter experts (SME), as described in Chapter 5) within their commands to:
 - (1) Integrate ESMRM requirements into the planning process and current plans to include all strategic lines of communication (LOC) nodes.
 - (2) Determine whether NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, can be met.
 - (3) Determine whether a munitions-related process requires either a national explosives safety site approval or an ESMRM risk assessment and risk decision as applicable. Figure 2-3 illustrates the national explosives safety site approval process² for a munitions-

² NATO does not have a NATO explosives safety site approval process. National site approval processes shall be used.

related process meeting NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable.

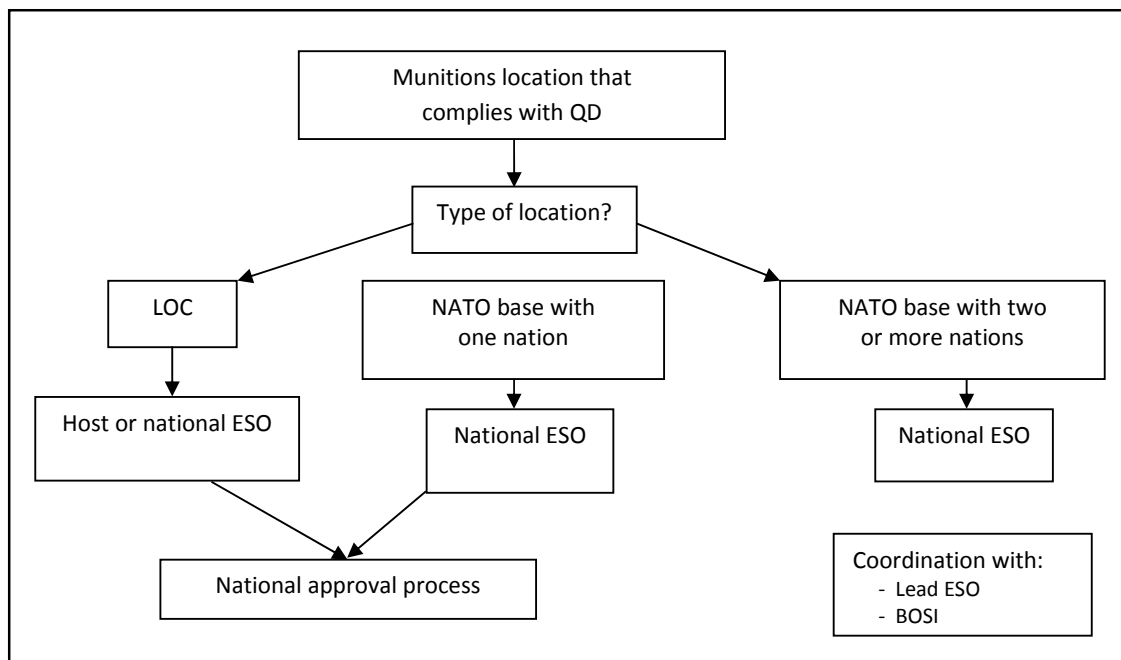


Figure 2-3. Explosives Safety Site Approval Process

- (4) Coordinate the development of ESMRM risk assessments when NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met.
- (5) Coordinate and communicate the risk decision at the delegated level within the NATO operational chain of command.
- (6) Implement and communicate risk decisions to all affected NATO command elements and nations.
- (7) Comply with the mitigating actions within the risk decision.
- (8) Maintain the risk assessment (i.e., validating whether conditions or plans require updated risk assessments).
- (9) Coordinate ESMRM risk assessment with other functional areas that have a stake in the process (e.g., force protection and security, construction engineers, military engineers, base operations).
- (10) Periodically review all existing risk decisions. Paragraph 7.4, Chapter 7, "Risk Assessment and Update Frequency," provides requirements for risk assessments.

- (11) Present previously approved JFC or component commander risk decisions to incoming JFC or component commander after a change of command for their risk decision.
- c. Direct the establishment and participation of nations in an ESB at all NATO bases with two or more nations present and where munitions are located and munitions-related processes occur.
- d. Direct the establishment of a Lead ESO position at all NATO bases with two or more nations present and where munitions are located and munitions-related processes occur.
- e. Direct the establishment of a BOSI at all NATO bases with two or more nations present and where munitions are located and munitions-related processes occur.
- f. Review and/or take action on all risk decisions for all locations where munitions are planned or present, to include logistics nodes and LOC, when NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met.
- g. Validate existing ESMRM risk decisions during the OPP.
- h. Maintain or have access to a repository of ESMRM risk assessments and risk decisions made within the area of operations (AOO).
- i. Coordinating and Communicating ESMRM:
 - (1) Coordinate with National Representatives. Coordinate all risk decisions with affected nations and obtain their concurrence with the ESMRM assessment, thereby informing affected nations of the associated munitions-related risks and encouraging their participation in the risk mitigation process³.
 - (2) Lack of Response from National Authorities. In the event that a nation chooses not to engage in the ESMRM process, NATO ESMRM requirements shall still apply and the ESMRM risk-decision shall be made by a general or flag officer delegated risk-decision authority within the NATO chain of command. The risk decision package shall document that a particular nation chose not to participate in the ESMRM process and their reason for doing so.

2.5. JTF (THEATER) COMMANDER ROLES AND RESPONSIBILITIES

2.5.1. NATO commanders shall integrate ESMRM across the full range of NATO military operations to include planning. Integrating ESMRM early in the planning

³ A national representative's concurrence on a risk assessment document indicates their awareness of their nation's exposure to munitions risk resulting from deviations from NATO explosives safety requirements and their participation in the risk mitigation process.

process and in the execution phases enables commanders to accomplish their missions and enhances mission effectiveness while minimizing potential exposure to the damaging consequences of munitions from accidental initiation or enemy fire.

2.5.2. All risk decisions involving munitions or munitions-related processes that do not meet NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, shall be based on ESMRM risk assessments as described in Chapter 7.

2.5.3. When delegated the ESMRM risk decision-making authority, NATO JTF commanders will make the risk decision within their chains of command as specified in paragraph 2.4.5. If not delegated the ESMRM risk decision-making authority, all ESMRM risk decisions will be forwarded to the JFC or component commanders.

2.5.4. Risk decisions affecting other nations' personnel, assets, or missions, as well as any unrelated personnel to the munitions operation, shall always be made by a general or flag officer within the NATO operational chain of command.

2.5.5. When delegated ESMRM risk decision-making authority, JTF commanders may further delegate ESMRM risk decision-making authority for the medium risk level identified in Table 7-2 in Chapter 7 to the senior MN base operational commander, when the senior MN base operational commander is a general or flag officer. The following apply:

- a. Risk assessments with a low risk level identified in Table 7-2 in Chapter 7 do not require approval delegation and can be approved by subordinate commanders.
- b. All risk decisions that would hazard other nations' personnel, assets, and un-related personnel must be communicated to and/or coordinated, as appropriate, with the affected nation.

2.5.6. Awareness of munitions-related risks shall be maintained by periodically reviewing all risk-decisions forwarded by subordinate commanders.

2.5.7. For the purposes of ALP-16, general staff (e.g., G/J-3, G/J-4, etc.) are considered part of the NATO operational chain of command.

2.5.8. JTF commanders shall:

- a. Make munitions-related risk decisions when delegated risk-decision authority following the process identified in subparagraph 2.4.5 and Chapter 7.
- b. Appoint ESMRM focal points (i.e., qualified explosives safety-related SME, as described in Chapter 5) within their commands to:
 - (1) Integrate ESMRM requirements into the planning process and current plans to include all strategic LOC nodes.

- (2) Determine whether NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, can be met.
 - (3) Coordinate the development of ESMRM risk assessments when NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met.
 - (4) Coordinate and communicate the risk decision at the delegated level within the NATO operational chain of command.
 - (5) Implement and communicate risk decisions to all affected NATO command elements and nations.
 - (6) Comply with the mitigating actions within the risk decision.
 - (7) Maintain the risk assessment (i.e., validating whether conditions or plans require updated risk assessments).
 - (8) Coordinate the ESMRM risk assessment with other functional areas that have a stake in the process (e.g., force protection and security, construction engineers, military engineers, base operations).
 - (9) Periodically review all existing risk decisions. Paragraph 7.4, Chapter 7, "Risk Assessment and Update Frequency," provides requirements for risk assessments.
 - (10) Present previously approved JTF commander risk decisions to incoming JTF commander after change of command for his/her risk decision.
- c. Direct the establishment and participation of nations in an ESB at all NATO bases with two or more nations present and where munitions are located and munitions-related processes occur.
 - d. Direct the establishment of a Lead ESO position at all NATO bases with two or more nations present and where munitions are located and munitions-related processes occur.
 - e. Direct the establishment of a BOSI at all NATO bases with two or more nations present and where munitions are located and munitions-related processes occur.
 - f. Review and/or take action on all risk decisions for all locations where munitions are planned or present, to include logistics nodes and LOC, when the requirements of AASTP-1 or AASTP-5, as applicable, cannot be met.
 - g. Maintain or have access to a repository of ESMRM-related risk assessments and risk decisions made within the AOO.

2.6. NATO BASE COMMANDER AND KEY NATO BASE ESMRM PERSONNEL ROLES AND RESPONSIBILITIES

2.6.1. Paragraph 2.6 applies to all forward operating bases (FOBs) and provides the NATO base commander's roles and responsibilities specific to the ESMRM process, and also provides roles and responsibilities for key ESMRM personnel on a base.

2.6.2. The NATO base commander and his ESMRM staff play a critical role in the ESMRM process at a NATO base. The NATO base commander has four primary responsibilities (i.e., 1. Oversight 2. Decision-Making 3. Implementing ESMRM Requirements and 4. Coordinate and Communicate ESMRM) that directly contribute to the management of ESMRM risks and required actions to identify, manage, and reduce those risks for all tenants on that NATO base:

- a. Oversight. Provide oversight of NATO base-related ESMRM programs to include those of nations having munitions and munitions-related processes on the base. ESMRM program details and critical explosives safety elements are identified in AASTP-1, AASTP-5, and ALP-16 (e.g., QD, FD, risk mitigation, munitions-related processes, handling, hazard classification, and surveillance). As part of their oversight responsibility, the NATO base commander shall:

- (1) Appoint or direct the appointment of key NATO base ESMRM personnel:

- (a) ESMRM qualified personnel. Direct NATO base tenant nations with munitions and munitions-related processes to assign ESMRM qualified personnel as prescribed in Chapter 5.

- (b) Lead ESO. Appoint a qualified Lead ESO if the NATO base has two or more nations present and where munitions are located and munitions-related processes occur. The Lead ESO:

- 1. Must meet the qualifications of an ESO as prescribed in Chapter 5.

- 2. Shall be a primary position and not a collateral duty.

- 3. Shall provide regular updates.

- (c) ESB Chair. The ESB chair shall have sufficient experience and rank to support the ESMRM program.

- (d) BOSI. The BOSI who performs other critical base operations support functions is a key element of an effective ESMRM program on the NATO base.

- (2) Integrate ESMRM requirements. Direct the integration of the following into NATO base master plans, NATO base maps, and other similar documents:

- (a) Approved national explosives safety site plans.
 - (b) Approved Risk decisions (and required mitigation actions).
 - (c) National ESO issued explosives licenses for their munitions locations on the NATO base.
- (3) Record Keeping. Direct the establishment and maintenance of an ESMRM continuity book as described below in subparagraphs 2.6.6.t and 2.7.3.q.
- b. Decision Making
 - (1) Serve as the final ESMRM decision maker, as necessary, to resolve munitions-related issues presented by the ESB chair or Lead ESO.
 - (2) Make medium-level ESMRM risk decisions when appropriate, and when delegated the authority to make those decisions.
 - (3) Make low-level ESMRM risk decisions, as necessary.
 - (4) Review for approval NATO base ESMRM operating instructions, memoranda of understanding (MOUs)/memoranda of agreement (MOAs), host/tenant accords, or other ESMRM-related agreements.
 - (5) Review and forward ESMRM risk decisions requiring higher level arbitration or approval to the appropriate commander within the NATO chain of command.
- c. Implementing NATO ESMRM Requirements. Require that all NATO base munitions locations and munitions-related functions meet the requirements of AASTP-1, AASTP-5, or ALP-16, as applicable. Specifically, each munitions location and munitions-related function shall have one of the following:
 - (1) A nationally approved explosives safety site plan.
 - (2) A national ESO issued explosives license.
 - (3) An approved ESMRM risk decision.
- d. Coordinating and Communicating ESMRM
 - (a) Establish an ESB. If the NATO base has two or more nations present and where munitions are located and munitions-related processes occur:
 - (a) Review for approval the ESB charter and terms of reference.

- (b) Require regular updates from the ESB chair.
- (b) Coordinate with National Representatives. Coordinate all risk decisions with affected nations and obtain concurrence with the ESMRM assessment thereby informing affected nations of the associated munitions-related risks and encouraging their participation in the risk mitigation process⁴.
- (c) Lack of Response from National Authorities. In the event that a nation chooses not to engage in the ESMRM process, NATO ESMRM requirements are still applicable and the ESMRM risk-decision shall be made by a general or flag officer delegated risk-decision authority within the NATO chain of command. The risk decision package shall document that a particular nation chose not to participate in the ESMRM process and their reason for doing so.

2.6.3. **Base Safety Officer.** The NATO base safety officer will:

- a. Develop and maintain a comprehensive NATO base ESMRM program.
- b. Enable access to NATO base leadership to communicate ESMRM information and issues.
- c. Request resources in support of ESMRM issues.
- d. Support the NATO base ESMRM mission by:
 - (1) Assigning and maintaining qualified ESMRM personnel on the NATO base.
 - (2) Supporting the NATO base ESB, Lead ESO, and the BOSI as required.

2.6.4. **Explosives Safety Board (ESB) Chair.** The ESB provides a framework for collectively resolving ESMRM and munitions-related issues on a NATO base. The ESB does not approve explosives safety site plans, explosives licenses, or ESMRM risk decisions. A detailed list of ESB responsibilities can be found in subparagraph 2.6.5. The ESB chair shall:

- a. In conjunction with the ESB, develop a charter and terms of reference for ESB membership and conduct of ESB activities.
- b. Update the NATO base commander regularly and present ESMRM issues when a decision is required.
- c. Regularly convene and preside over the NATO base ESB meetings.

⁴ A national representative's concurrence on a risk assessment document indicates their awareness of their nation's exposure to munitions risk resulting from deviations from NATO explosives safety requirements and their participation in the risk mitigation process

2.6.5. Explosives Safety Board (ESB). The ESB, which is typically comprised of the ESB chair, Lead ESO, BOSI, tenant ESMRM and munitions representatives, base operations, master planning, representatives of nations that are encumbered (i.e., exposed to munitions risk), and others as required (see Figure 2-2), shall:

- a. Based on the scope and types of operations being performed, determine, based on the Lead ESO recommendation, whether NATO AASTP-1 or AASTP-5 is more appropriate for use at the NATO base.
- b. Address ESMRM issues presented by the Lead ESO. Typically, ESB business will revolve around critical explosives issues pertaining to encroachment inside or near quantity distance (QD) as defined in AASTP-1 and field distance (FD) as defined in AASTP-5. AASTP-5 provides a list of critical explosives safety elements that must be monitored and managed on a NATO base for a comprehensive, adequate, and safe ESMRM program. Most ESB issues will fall in the area of real estate and facilities usage or placement; construction management; conflicting or competing interests for land, facilities, or functions; issues with national participation; and sharing of information.
- c. Resolve, whenever possible, munitions-related issues presented to the Board.
- d. Develop potential solutions (for issues identified but not limited to those above) for the NATO base commander's decision when not able to resolve munitions-related issues.
- e. Record and distribute meeting minutes, to include NATO base commander endorsement of ESB decisions and approvals of ESB recommendations, to all potentially affected parties of those decisions.

2.6.6. Lead ESO. The Lead ESO manages and provides oversight of the NATO base commander's ESMRM program and acts as the focal point for all ESMRM-related activities on the NATO base, including those of tenant commands on the NATO base. The Lead ESO shall:

- a. Advise the NATO base commander on ESMRM issues and provide recommendations, as necessary, on appropriate actions to take.
- b. Participate in the ESB.
- c. Prepare for the NATO base commander's approval, base ESMRM operating instructions, MOUs/MOAs, host/tenant accords, or other ESMRM-related agreements.
- d. Provide ESMRM oversight of all munitions-related functions on the NATO base.

- e. Recommend to the ESB whether AASTP-1 or AASTP-5 is more appropriate to apply to munitions locations and munitions-related functions on the NATO base. The recommendation could change if the NATO base and its supporting infrastructure expands and develops into a more substantial/permanent NATO base.
- f. Develop and maintain a list of all key ESMRM-related personnel on the NATO base to include tenants. The Lead ESO shall also develop and maintain an awareness of the ESMRM personnel within the NATO chain of command.
- g. Identify ESMRM and munitions-related issues to the ESB chair, as necessary, for ESB involvement to include potential QD or FD encroachment issues both internal and external to the base boundary.
- h. Develop and maintain a working relationship with all ESMRM-related personnel on the NATO base (e.g., BOSI, nations' ESOs, and base operational support organizations/ functions). The primary focus shall be to comply with NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, and resolve any ESMRM issues at the lowest possible command structure level.
- i. Meet with the leadership of all new NATO base tenants immediately upon their arrival and do the following:
 - (1) Inform them of NATO base ESMRM requirements and provide them with copies of all NATO base ESMRM-related documents.
 - (2) If they have/will have munitions at the NATO base:
 - (a) Obtain contact details for national ESMRM personnel.
 - (b) Assist national ESMRM personnel in integrating their munitions and explosives-related operations and processes into the NATO base ESMRM process.
- j. Serve as the primary explosives safety interface between nations and BOSI with respect to NATO base QD or FD maps, real estate needs, construction management, use of facilities, etc. The Lead ESO shall maintain situational awareness of all munitions and munitions-related processes on the base to include activities occurring or planned within or affected by QD or FD.
- k. Coordinate ESMRM requirements with base support functions (e.g., BOSI, security, force protection, military engineers) that have overlapping areas of concerns related to munitions hazards/threats and mitigation and develop mutually acceptable solutions as required.
- l. Participate in all NATO base planning meetings to maintain visibility of all ESMRM issues and reduce the likelihood of potential problems.

- m. Review and endorse all nationally-developed explosives safety site plans prior to their submission for approval through national processes. Figure 2-3 describes the NATO base explosives safety site approval process.
- n. Coordinate the integration of QD or FD, from approved national site plans, risk decisions, and explosives licenses into NATO base master planning, NATO base maps, and other NATO Base planning and operational documents, as appropriate.
- o. Monitor compliance with the conditions contained in approved explosives safety site plans, risk decisions, and explosives licenses.
- p. Support the preparation of risk assessments, as described in Chapter 7, when explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met, and actively participate in risk mitigation to the maximum extent practicable.
- q. Monitor all risk assessments and decisions for currency. Paragraph 7.4, Chapter 7, "Risk Assessment and Update Frequency," provides requirements for risk assessment maintenance.
- r. Conduct a review of each nation's ESMRM program upon their arrival and, at a minimum, every 6 months thereafter for compliance with base standard operating procedures (SOPs), instructions, MOUs/MOAs, or other ESMRM governing documents.
- s. Following a review, provide a written report to the reviewed nation's commander and ESO and the base commander. Significant findings that pose an immediate risk shall be immediately reported to the base safety officer and NATO base commander and must be resolved without delay.
- t. Develop and maintain an ESMRM continuity binder that shall be handed over to the next Lead ESO. Contents shall include:
 - (1) The most current NATO base map showing munitions-related locations (current and planned) and associated QD or FD
 - (2) NATO base ESMRM-related SOPs, instructions, and MOUs/MOAs
 - (3) List of key NATO base national ESMRM personnel and their contact information
 - (4) Copies of all nationally-issued explosives licenses
 - (5) Copies of all nationally-issued explosives safety site plans
 - (6) Copies of Lead ESO review reports and responses to findings

- (7) List of ESMRM key personnel in the NATO, NATO base, and national chains of command (see Figure 2-2)
- (8) Copies of ESMRM risk assessments documentation for the NATO base and its tenants, to include those provided to the chain of command
- (9) Other pertinent information

2.6.7. **BOSI.** Although the BOSI performs a variety of other NATO base support functions, the BOSI is a key ESMRM individual who helps implement and manage an effective ESMRM program on the NATO base. The BOSI shall:

- a. Support the Lead ESO and ESB chair on NATO base ESMRM matters.
- b. Participate in the ESB.
- c. Participate in all planning meetings relating to munitions locations and munitions-related processes and other projects that fall within the QD or FD.
- d. Monitor and communicate to the Lead ESO existing and potential QD or FD encroachment issues both internal and external to the base boundary.
- e. Maintain awareness of all proposed and approved site plans as they could potentially impact master planning documents.
- f. Coordinate the integration of the QD or FD from approved national site plans, risk decisions, and explosives licenses into the NATO base master planning process (including base maps) and applicable planning documents.
- g. Coordinate with the Lead ESO regarding NATO base real estate issues, new or proposed construction, new or proposed functions/operations contracts, etc., within the QD or FD arcs.
- h. Support, as necessary, the conduct of ESMRM risk assessments for NATO base munitions locations and munitions-related processes that cannot meet the NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable.

2.7. **NATO OPERATIONAL COMMANDER AND KEY NATIONAL ESMRM PERSONNEL ROLES AND RESPONSIBILITIES (AS A TENANT ON A NATO BASE)**

2.7.1. Paragraph 2.7 applies to all FOBs and provides the NATO operational commander's roles and responsibilities specific to their and the NATO base's ESMRM processes. It also provides roles and responsibilities for key national personnel involved in the ESMRM process as a tenant on a NATO base.

2.7.2. NATO Operational Commander. Tenant nations on a NATO base having a munitions-related mission or supporting munitions-related processes play a critical role in the ESMRM process (see Figure 2-2). The operational commander's four primary responsibilities, detailed below, directly support the NATO base commander's ESMRM program and contribute to the overall management of ESMRM risks and required actions to identify, manage, and reduce those risks for all tenants on that base:

- a. Oversight. Provide oversight of the ESMRM program for their national munitions and munitions-related processes on the NATO base. ESMRM program details and critical explosives safety elements are identified in AASTP-1, AASTP-5, and ALP-16 (e.g., QD, FD, risk mitigation, munitions-related processes, handling, hazard classification, and surveillance). Operational commanders shall:
 - (1) Direct their national support personnel, identified in Figure 2-2, to comply with NATO base ESMRM requirements.
 - (2) Appoint a qualified national ESO for munitions and munitions-related processes. The individual assigned to the national ESO position must meet the qualifications of an ESO, as prescribed in Chapter 5.
 - (3) Direct their national ESO to:
 - (a) Monitor their national munitions locations and munitions-related processes on the NATO base.
 - (b) Coordinate with the Lead ESO and prepare national ESMRM explosives safety site plans for munitions locations and munitions-related processes on the NATO base.
 - (c) Coordinate with the Lead ESO and affected nations and prepare risk assessments, as prescribed in Chapter 7, in support of national risk decisions (and required mitigation actions), when NATO explosives safety requirements, as established by the ESB at the NATO base, cannot be met.
 - (d) Provide copies of nationally approved explosives safety site plans, risk assessments, and decisions to the Lead ESO.
 - (e) Participate in the ESB and support the ESB chair as required.
 - (f) Issue explosives licenses for all national munitions locations and munitions-related processes. Such licenses shall be based on a nationally approved explosives safety site plan, an approved risk decision, or other national approval document.
 - (g) Participate in risk assessments and risk mitigation resolution efforts when your nation is potentially impacted by explosives-related risk from another nation at the NATO base.

- (h) Periodically validate that conditions contained in explosives licenses and risk decisions continue to be met.
- (i) Direct the establishment and maintenance of a national ESMRM continuity book as described below in subparagraph 2.7.3.q.

b. Decision Making

- (1) Serve as the national ESMRM decision-maker, as necessary, to resolve munitions-related issues presented by the national ESO.
- (2) Make medium ESMRM risk decisions, when delegated the authority to make those decisions, and low risk decisions.
- (3) Review and forward ESMRM risk decision issues requiring higher level arbitration or approval to the base commander.
- (4) Review for national endorsement base ESMRM operating instructions, MOUs/MOAs, host/tenant accords, or other ESMRM-related agreements.
- (5) Review for endorsement national explosives safety site plans and forward to appropriate national approval authority.

c. Implementing NATO ESMRM Requirements. Require that all national munitions locations and munitions-related functions on the base meet the requirements of AASTP-1, AASTP-5, or ALP-16, as applicable. Specifically, each munitions location and munitions-related function shall have:

- (1) A national approval (i.e., explosives safety site plan),
- (2) A national ESO issued explosives license, or
- (3) An ESMRM risk decision.

d. Coordinating and Communicating ESMRM

- (1) Coordinate medium ESMRM risk decisions, when delegated risk decision-making authority, and low risk decisions with affected nations.
- (2) Communicate the associated munitions-related risks to affected nations and participate in the risk mitigation process.

- (3) Endorse, as recommended by their national ESO, risk assessment conducted by other nations for explosives operations which affect your national personnel and assets.⁵

2.7.3. National ESO. The national ESO shall:

- a. Be familiar with all NATO base ESMRM requirements and related documents.
- b. Comply with the NATO base ESMRM requirements.
- c. Serve as the focal point for all national ESMRM-related activities on the NATO base.
- d. Manage their nation's ESMRM program on behalf of their national operational commander.
- e. Support the ESB chair, Lead ESO, and BOSI in their NATO base ESMRM oversight responsibilities.
- f. Participate in the NATO base ESB process.
- g. Prepare explosives safety site plans for approval through their national process for munitions operations that can meet the ESMRM requirements of AASTP-1 or AASTP-5, as applicable. Figure 2-3 describes the explosives safety site approval process.
 - (1) Coordinate explosives safety site plans with the Lead ESO and the BOSI during development.
 - (2) Communicate approved QD or FD to the Lead ESO and BOSI and provide them copies of nationally-approved explosives site plan documents.
- h. Identify national ESMRM and munitions-related issues to the Lead ESO for coordination with the ESB chair, as necessary.
- i. Maintain a working relationship with all ESMRM-related personnel on the NATO base (e.g., Lead ESO, other nations' ESOs, BOSI, and base operational support organizations/ functions).
- j. Serve as the ESMRM interface between their nation and the Lead ESO with respect to NATO base QD or FD maps, real estate needs, construction management, use of facilities, etc. The national ESO shall maintain awareness of all national activities occurring or planned within the QD or FD from their munitions locations and munitions-related processes on the NATO base.

⁵ A national representative's concurrence on a risk assessment document indicates their awareness of their nation's exposure to munitions risk resulting from deviations from NATO explosives safety requirements and their participation in the risk mitigation process

- k. Participate in all NATO base planning meetings involving national ESMRM areas to maintain total visibility of all ESMRM issues and help avoid potential problems.
- l. Prepare risk assessments, as described in Chapter 7, when explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met, and mitigate risk to the maximum extent practicable.
- m. Participate in risk assessments being conducted by another nation when their national personnel and assets are affected by the other nation's explosives risks and assist in mitigating those risks to the extent possible.
- n. Make recommendations to the national operational commander about endorsing another nation's risk assessment when affected by explosives-related risk from that nation. The affected nation should participate in the risk assessment and risk mitigation effort to the maximum extent possible.
- o. Conduct inspections of their nations' munitions locations and munitions-related processes.
- p. Document findings from inspections and inform the national commander. Any noncompliance to NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, shall be reported to the Lead ESO. Significant findings that pose an immediate risk shall be immediately reported to the national commander and must be resolved without delay.
- q. Develop and maintain an ESMRM continuity binder that shall be handed over to the next national ESO. Contents shall include:
 - (1) Current NATO base map showing munitions-related locations (current and planned) and the associated QD or FD.
 - (2) NATO base and national ESMRM-related SOPs, instructions, and MOUs/MOAs.
 - (3) List of key national and NATO base ESMRM personnel and their contact information (See Figure 2-2).
 - (4) Copies of all national ESO-issued explosives licenses.
 - (5) Copies of nationally approved explosives safety site plans.
 - (6) Copies of national ESO inspection/review reports and responses to findings.
 - (7) Copies of national ESMRM risk assessments documentation provided to the chain of command.
 - (8) Other pertinent information.

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CHAPTER 3 INTEGRATING EXPLOSIVES SAFETY AND MUNITIONS RISK MANAGEMENT (ESMRM) INTO NATO PLANNING AND PREPARATION

3.1. PURPOSE

The purpose of this chapter is to establish and describe ESMRM process requirements in relation to the NATO Operational Planning Process across the full range of NATO military operations. The information contained in Chapter 3 augments Chapter 4, “Logistic Planning and Preparation,” in Allied Logistic Publication 4.2(A), “Land Forces Logistic Doctrine,” by providing ESMRM-specific requirements.

3.2. PLANNING AND ESMRM

3.2.1 Integrating ESMRM requirements into the NATO Operational Planning Process (OPP). Planning involves critical decisions concerning the interface of combat, combat support, and combat service support (CSS) at all levels. ESMRM requirements, as part of Logistics Planning and subsequent operations, must be versatile, dedicated to the aim of the mission, and fully integrated within all phases of the combat planning process. At the operational and tactical levels, ESMRM links operations to the Consumer Logistics Process phases identified in Chapter 4. Figure 3-1 illustrates this relationship.

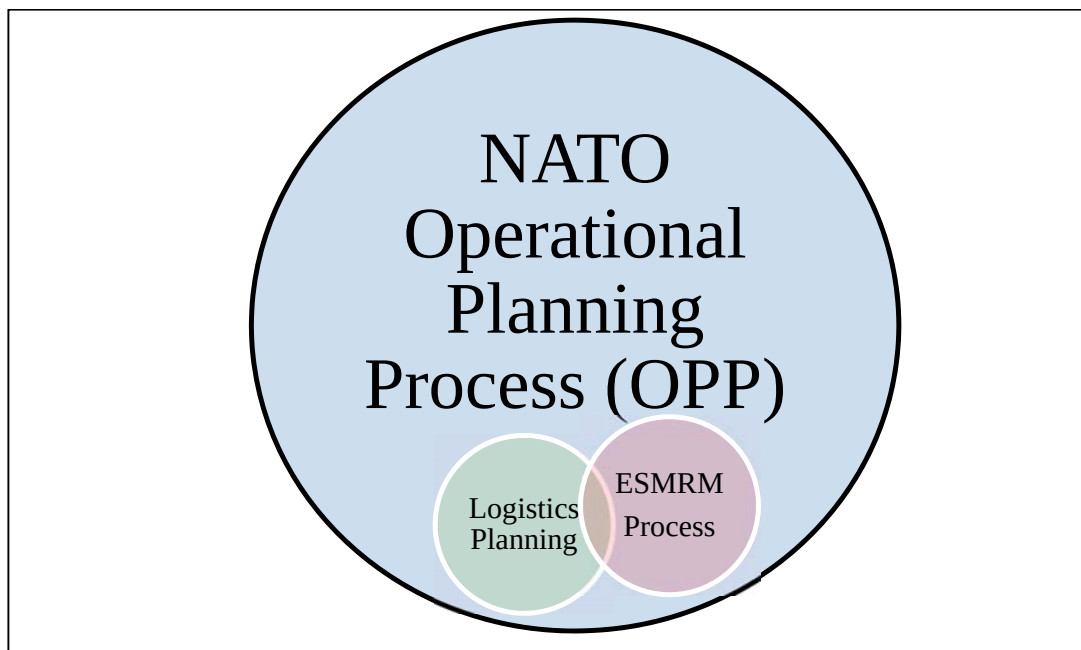


Figure 3-1. Overlap of ESMRM within the NATO Logistics Planning Process and the OPP

3.2.2. Mission Support. The ESMRM process, when integrated into the Logistics Planning process within the OPP, supports the mission and subsequent operations

by identifying munitions-related risks in support of the NATO commander's risk decision within all phases of the combat planning process to include combat, combat support, and CSS.

3.2.3. NATO OPP. When operations and plans involve munitions, ESMRM must be fully integrated into the Logistics Planning and OPP. To succeed, ESMRM must be integrated with, and must support, the CSS, tactical, and operational plans. Staff dealing with ESMRM must be brought into the planning process at the outset since conditions involving munitions will very likely have a profound effect on an emerging plan. Furthermore, it is vital throughout the OPP and the issuing of all the attendant authorizations, that ESMRM be incorporated in the logistics estimate and developed in parallel. Any nation where munitions are planned to be received, transported, stored, handled, maintained, employed, or disposed of must be included and involved throughout the process (i.e., strategic, tactical, and operational levels).

3.2.4. Operation Plan (OPLAN). ESMRM risk decisions must be part of the logistics concept of operations in the final OPLAN and cannot be written or coordinated without national involvement.

3.3. ESMRM AS PART OF THE LOGISTICS PLANNING PROCESS

3.3.1. Phases of the OPP. Allied Command Operation (ACO) "Comprehensive Operations Planning Directive (COPD)" is the basic guide for NATO's OPP development by NATO strategic, regional, and sub-regional commanders both for Article 5 and non-Article 5 crisis response operations. This Section addresses how ESMRM integrates into the five Logistics Planning phases: initiation, orientation, concept development, plan development and plans review. ESMRM personnel shall focus on risks to and from munitions throughout the Logistics Planning process. Additionally, the five phases are described in the NATO OPP Development Guide.

3.3.2. Initiation. OPP initiation is a NATO response to an event that may influence NATO commanders in an area of operations (AOO) and area of interest (AOI). Initiation is only the commencement of planning for an operation and not the execution of any operation. Before issuing an initiating directive (ID), advice concerning the potential use of munitions should be requested. After a determination of munitions involvement, ESMRM personnel (as described in Chapter 2) shall support logistics planners. ESMRM personnel must be aware of events and any relevant previously developed contingency operation plan (COP) or OPLAN involving munitions. In the initiation phase, ESMRM personnel shall:

- a. Support the planning staff by maintaining awareness of the situation since planning will only commence once the ID from the higher headquarters (HQ) is received. In anticipation of the ID, ESMRM personnel shall develop a potential list of involved nations and potential lines of communication (LOC) (e.g., commercial air/seaports, railheads, and military installations) that could be used in support of the mission in response to any ID received.
- b. Identify to the planning staff coordinator any significant ESMRM concerns that could affect NATO operations in the AOO/AOI in anticipation of the ID.

- c. Upon receipt of the ID, support the planning team that will be formed and provide required ESMRM information to logistics planners to support the commander's plan.

3.3.3. Orientation. The orientation phase begins with the receipt of the ID and results in the issuing of the NATO Commander's Planning Guidance. ESMRM personnel will provide logistics planners with ESMRM information in support of the mission analysis process and provide assistance in the development of the Mission Analysis Brief and formation of the NATO Commander's Planning Guidance. In the orientation phase, ESMRM personnel as part of the planning staff shall:

- a. Participate in the review and analysis of the mission statement from the higher HQ to identify and address risks to and from munitions.
- b. Maintain awareness through analysis of the ID and intelligence information.
- c. Maintain awareness of the NATO commander's intent through analysis of the commander's guidance.
- d. Maintain awareness of the mission objectives through analysis of the planning factors. Assess the feasibility of supporting the force from an ESMRM perspective.
- e. Know the mission end-states and analyze the logistics of the situation(s).
- f. Know the restraints and constraints of the planning guidance and the potential impact to providing support.
- g. Identify any Logistics Planning assumptions for courses of action (COAs) development that require consideration for continued plan development.
- h. Know and understand the centers of gravity and how risks to and from munitions affect them.

3.3.4. Concept Development. Review of the NATO Commander's Planning Guidance during the Concept Development phase provides guidance and direction to the planning staff to develop the COA. ESMRM personnel support the planning staff in the COA development and concept of resulting operations. The critical action in the concept development phase is the staff assessment of each COA. ESMRM personnel shall analyze proposed COAs to determine the viability and sustainability of each COA and provide recommendations and rankings of the COAs from a logistics perspective focusing on risks to and from munitions. The concept development phase will produce the Concept of Operations Decision Briefing and the commander's selection of a COA for further refinement and development. In the concept development phase, ESMRM personnel shall:

- a. Assist in the COA development process by identifying and assessing risks associated with munitions and munitions-related support (i.e., Consumer Logistics Process described in Chapter 4) operations.

- b. Make recommendations concerning COA selection based on risk from required munitions and munitions-related support operations.
- c. Provide, as part of the planning staff, ESMRM input for the decision briefing to the NATO commander.
- d. Participate, upon the NATO commander's approval of a COA, in the development of the OPLAN.

3.3.5. Plan Development. During the Plan Development phase, planners develop and coordinate the desired plan, seek its approval, and issue the plan. ESMRM personnel support logistics planners who, in turn, support the operational planners in the development of the plan based on the NATO commander's selected COA. ESMRM personnel will prepare an ESMRM risk assessment that will be included in the report discussed in Chapter 7. An integral part of the ESMRM process is the determination of whether NATO explosives safety requirements¹ can be met. In the event that these requirements cannot be met, a NATO commander's risk decision will be required for each stage/location. The ESMRM appendix shall contain the appropriate risk assessments per Chapters 4 and 7, as well as the NATO commander's risk decisions. Although the final output in the Plan Development phase is the NATO commander's approved plan, risk decisions for required munitions and munitions-related support operations at strategic LOC are a critical part of the COP or OPLAN. In the plan development phase, ESMRM personnel shall:

- a. Work on development of the OPLAN and the ESMRM risk assessments for the selected COA.
- b. Maintain situational awareness of force capability requirements for combat support and CSS units to understand required munitions and munitions-related support operations. Maintaining situational awareness can require liaising with subordinate HQ such as joint forces and component commands.
- c. Maintain situational awareness for the planning of the force deployment, in accordance with the principles of reception, staging, onward movement and integration (RSOI) as defined in Allied Joint Publication (AJP) 3.13, "Allied Joint Doctrine for the Deployment of Forces," and AJP-4.4, "Allied Joint Movement and Transportation Doctrine."
- d. Maintain situational awareness of the sustaining requirements for force deployment that address munitions and munitions-related support operations.

¹ Allied Ammunition Storage and Transport Publication (AASTP)-1, "Manual of NATO Safety Principles for Storage of Military Ammunition and Explosives," and AASTP-5, "NATO Guidelines for the Storage, Maintenance and Transport of Ammunition on Deployed Missions or Operations"

- e. Develop the required ESMRM sections for munitions and munitions-related support operations as part of the NATO base planning document. Real estate requirements shall be addressed in the NATO base planning document.
- f. ESMRM requirements shall be included in existing NATO plans as a separate annex or as part of an existing section.

3.3.6. Plans Review. The plans review phase includes the plan review and plan evaluation steps. Plans are continually reviewed and analyzed to minimize potential negative impacts and if required, a new decision brief may be required. The overall product of the plans review phase is the continued applicability of the plan. ESMRM personnel shall actively participate in the periodic review of developed plans and update/revise them and supporting documents as required.

3.4. ESMRM AS PART OF LOGISTICS PLANNING CONFERENCES

3.4.1. Logistics Planning Conferences. Logistics Planning Conferences are integral to achieving unity of effort. NATO HQ, staffs, and nations share the responsibility for the planning process and planning can be facilitated through Logistics Planning Conferences.

3.4.2. ESMRM Personnel Participating in Logistics Planning Conferences. ESMRM personnel shall participate in Logistics Planning Conferences to determine munitions and munitions-related support operations requirements in support of:

- a. ESMRM risk assessments described in Chapter 7.
- b. NATO Base Planning Documents in support of the OPLAN.

3.4.3. Logistics Planning Conferences

a. Initial Logistics Planning Conference

- (1) Purpose: Inform nations of the mission and concept of operations, analyze factors influencing Logistics Planning, adjust principles of operation, refine the logistics concept, and review the logistics structure and command and control.
- (2) Result: Enable strategic commands (SCs) to develop the logistics annex to OPLAN and provide information so that national Logistics Planning can begin.

b. Main Logistics Planning Conference

- (1) Purpose: Explain and discuss the OPLAN; identify logistics requirements, common logistics functions and procedures, host nation support (HNS), and legal and funding issues; and commence the logistics force planning process.

- (2) Result: Operational commands can finalize OPLAN and develop detailed force requirements; SC/operational commands can initiate the HNS process and nations can develop and detail logistics plans.

c. Operations and Logistics Review Conference (Optional)

- (1) Purpose: The Operations and Logistics Review Conference is held only if changes in the military situation have occurred that would affect Logistics Planning, or if identified shortfalls could prevent mission accomplishment.
- (2) Result: An executable OPLAN.

d. Final Logistics Planning Conference (FLPC)

- (1) Purpose: Finalize Logistics Planning, optimize overall logistics support, confirm command and control structure, and resolve remaining logistics issues.
- (2) Result: A balanced and harmonized system of SC, operational commands, and national logistics plans.

e. Movement & Transport (M&T) Planning Conferences and Medical Planning Conferences. Details on M&T planning and medical planning are contained in AJP-4.4 and AJP-4.10, "Allied Joint Medical Support Doctrine."

3.4.4. **ESMRM Planning for Concluding the Operation.** Historically munitions retrograde and disposal have not been considered during planning. Munitions and munitions-related support operations shall be included in ESMRM risk assessment described in Chapters 6 and 7, as well as the NATO Base Planning Document in support of the OPLAN.

3.5. ESMRM AS PART OF THE PREPARATION

3.5.1. **Logistics Preparation of the Theatre (LPT).** LPT is a key tool available to NATO commanders and their planners in building a flexible operational support plan. LPT consists of actions taken by logisticians at all echelons to optimize means (e.g., force structure, resources, strategic lift) of logistics support of the NATO commander's plan. Any ESMRM decisions made during the planning process shall be included in the LPT. Since ESMRM and munitions management are specific skills within the logistics area, ESMRM personnel shall be consulted to provide critical input to logisticians during the LPT.

3.5.2. **ESMRM as Part of Logisticians' Actions in Support of the LPT.** The LPT is a living document that will be in a continual state of review, refinement, and use. Logisticians' actions focus on identifying the resources currently available in the theatre for use by friendly forces and planning for access to resources. ESMRM information is key to logisticians and integral to follow-on decisions and actions, which include:

- a. Identifying and preparing Forward Operating Bases (FOBs). ESMRM plays a vital role in effectively managing a FOB since munitions areas must be planned for before the NATO base is established. All munitions-logistics functions are likely to take place at a FOB and should be considered during the planning process. Examples include but are not limited to:
 - (1) Reception.
 - (2) Storage. In addition to planning for NATO operating forces, logisticians should plan for captured enemy ammunition (CEA) as part of the munitions storage area since CEA has additional requirements.
 - (3) Munitions transportation routes in and around the NATO base.
 - (4) Distribution.
 - (5) Maintenance areas when munitions and/or weapons systems are involved.
 - (6) Munitions loading and unloading and turn-in areas for personnel and vehicles (including aircraft) when returning from training and/or combat operations.
 - (7) Demilitarization of munitions and/or weapons systems.
- b. Selecting and improving the LOC. ESMRM information is critical in the decision to select the LOC since risks to and from munitions during the RSOI phase that do not meet NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable. When NATO explosives safety requirements cannot be met at the LOC, the NATO commander shall make the ESMRM risk decision.
- c. Projecting and preparing forward logistics bases (FLBs). FLBs shall adhere to the same requirement as those requirements identified for FOBs when munitions and weapons systems are involved.
- d. Forecasting and building operational stock assets forward and in reserve. When munitions and weapons systems are involved, NATO commanders shall make every attempt to comply with the requirements of NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable. When these requirements cannot be met, the ESMRM process (i.e., assessment in preparation for the commander's risk decision and the risk decision) shall be followed as prescribed in ALP-16.

3.5.3. ESMRM as part of the Logistics Information Database. Once a contingency country or geographic region is known, logistics planners must begin to build a logistics information database. ESMRM risk decisions shall be included in the Logistics Information Database and include required risk mitigating action and

quantity-distance or field distance limitations. When completed, the information on the database can be used to develop a comprehensive plan for LPT.

3.5.4. LPT Elements of Information. The focus of the logistics elements of the information development process is on supply and field services. A detailed LPT plan shall cover all logistics areas to include ESMRM information and shall address the seven phases of the Consumer Logistics Process identified in Chapter 4.

3.6. ESMRM AS PART OF THE PLANNING FOR CONCLUDING A MILITARY OPERATION

3.6.1. Historically, nations involved in operations have not focused on the ESMRM requirements to successfully conclude a military operation; therefore, long-term, negative consequences have resulted. Failure to plan for and conduct an orderly end to the operation will endanger forces, national civilian populations, and the environment and may result in the loss of resources (e.g., money and equipment), life, morale, public support, and goodwill. Concluding a military operation and redeployment has two significant aspects for munitions: retrograde movement and in-theatre disposal, especially destruction.

3.6.2. Planning for ESMRM as part of the overall operation conclusion shall be part of both the OPP and Logistics Planning processes.

3.6.3. A Munitions Retrograde Plan shall be developed as part of the OPP and refined throughout the duration of the operation. The plan, at a minimum, shall address:

- a. Host nation pertinent information (e.g., port capabilities, other infrastructure information, challenges).
- b. Types and quantities of munitions.
- c. Locations for redeployment (e.g., FOBs and FLBs, intermediate storage, over-the-road terminals, railheads, air or seaports).
- d. Any intermediate temporary holding/storage locations.
- e. Means of transport both within and out of theatre.
- f. Additional transportation requirements.
- g. Shipping and receiving nations.
- h. Required packaging.
- i. Munitions hazard classification.
- j. Packaging configurations involved (e.g., break bulk, containerized).

- k. Material handling equipment/weight handling equipment and associated handling gear.
 - l. Special considerations as necessary, including but not limited to:
 - (1) Depleted uranium handling.
 - (2) Damaged weapons systems, vehicles, or equipment suspected to contain munitions.
 - (3) Captured enemy ammunition identified for destruction or retrograde.
 - (4) Means to acquire and repackage munitions returned without packaging.
 - m. Means and potential locations for demilitarization.
 - n. Personnel requirements for all parts of the retrograde plan (e.g., munitions handlers, munitions inspectors, munitions transporters, required qualifications, ESMRM personnel, explosive ordnance disposal, transport of dangerous/ hazardous materials shipping specialists).
 - o. Post-operation activities. The plan shall include ESMRM aspects associated with munitions-related processes and activities:
 - (1) The potential transfer or destruction of any munitions operations infrastructure built by NATO member nations in support of the mission.
 - (2) Removal of explosive remnants of war.
 - (3) Closing or transferring training ranges.
 - (4) Any additional ESMRM issues not identified in subparagraph 3.6.3.
 - p. Technical munitions and ESMRM personnel advanced party in preparation for the conclusion phase.
- 3.6.4. ESMRM personnel shall evaluate the Munitions Retrograde Plan during the operational phase to determine likely areas requiring ESMRM risk assessments and risk decisions when applicable parts of AASTP-1 or AASTP-5 requirements cannot be met.
- 3.6.5. Chapter 6 provides ESMRM requirements for concluding a military operation.

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CHAPTER 4 INTEGRATING EXPLOSIVES SAFETY AND MUNITIONS RISK MANAGEMENT (ESMRM) INTO THE NATO CONSUMER LOGISTICS PROCESS ACROSS THE FULL RANGE OF NATO MILITARY OPERATIONS
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4.1. PURPOSE

The purpose of this chapter is to identify the seven ESMRM risk assessment types and link them to the seven phases of the NATO Consumer Logistics Process across the full range of NATO military operations.

**4.2. NATO CONSUMER LOGISTICS PROCESS PHASES AND ASSOCIATED
ESMRM RISK ASSESSMENT TYPES**

4.2.1. NATO Consumer Logistics Process Phases. Reducing munitions-related risks requires integrating ESMRM tenets and requirements throughout the full range of NATO military operations. Figure 4-1 illustrates the seven phases of the NATO Consumer Logistics Process as detailed in Allied Logistic Publication (ALP)-4.2(A), "Land Forces Logistic Doctrine." Figure 4-2 illustrates the relationship between the NATO Operational Stages, the NATO Consumer Logistics Process phases, and the ESMRM applicability across the full range of NATO military operations. Logistics process phases described herein, and shown in Figure 4-1, may occur independently of each other therefore flexibility should be used when applying ESMRM to those phases.

4.2.2. NATO Consumer Logistics Process Phases and Associated ESMRM Risk Assessments Types. Table 4-1 lists the seven types of ESMRM risk assessments and links them to the NATO Operational Stages.



Figure 4-1. Seven Phases of NATO Consumer Logistics



*RSOI (Reception, staging, onward movement and integration)

Figure 4-2. Relationship of ESMRM Process to the NATO Operational Stages and the Consumer Logistics Process

Table 4.1. NATO Crisis Response Planning* and Associated ESMRM Risk Assessment Type

NATO Consumer Logistics Process Phase	ESMRM Risk Assessment Type	ALP-16 Reference	NATO Operational Stages
Reception (includes collection and consolidation during disestablishment)	RSOI	Chapter 4 Paragraph 4.3 and Chapter 6 paragraphs 6.4 and 6.11	Deployment and Disestablishment
Storage	Storage	Chapter 4 Paragraph 4.4 and Chapter 6 paragraphs 6.5 and 6.11	Employment, Redeployment, and Disestablishment
Transportation	Transportation	Chapter 4 Paragraph 4.5 and Chapter 6 paragraphs 6.6 and 6.11	Deployment, Employment, Redeployment, and Disestablishment
Distribution (includes collection and consolidation during disestablishment)	Distribution	Chapter 4 Paragraph 4.6 and Chapter 6 paragraphs 6.7 and 6.11	Employment and Disestablishment
Maintenance (includes preparation during disestablishment)	Maintenance and Handling	Chapter 4 Paragraph 4.7 and Chapter 6 paragraphs 6.8 and 6.11	Employment and Disestablishment
Retrograde	Retrograde and Removal	Chapter 4 Paragraph 4.8 and Chapter 6 paragraphs 6.9 and 6.11	Redeployment and Disestablishment
Disposal	Disposal and Demilitarization	Chapter 4 Paragraph 4.9 and Chapter 6 paragraphs 6.10 and 6.11	Redeployment and Disestablishment

*Allied Joint Publication (AJP)-5, "Allied Joint Doctrine for Operational-Level Planning," Figure 3.6, page 3-63.

4.3. RECEPTION

4.3.1. Reception. Reception generally occurs during reception, staging, onward movement and integration (RSOI)¹. Reception generally takes place at a line of

¹ (RSOI) as defined in Allied Joint Publication (AJP) 3.13, "Allied Joint Doctrine for the Deployment of Forces," and AJP-4.4, "Allied Joint Movement and Transportation Doctrine."

communication (LOC) node where munitions are unloaded from a means of transport in support of an operation, exercise, or other NATO mission. Ammunition Movement Publication (AMovP)-6, "Allied Multimodal Transportation of Dangerous Goods Directive," provides specific requirements for the movement of dangerous goods, to include munitions.

4.3.2. ESMRM RSOI Risk Assessment. Typically, RSOI of munitions occurs at commercial locations (e.g., ports, airfields, railheads) that are extremely congested and frequently cannot meet required NATO explosives safety requirements given in Allied Ammunition Storage and Transport Publication (AASTP)-1, "Manual of Safety Principles for the Storage of Military Ammunition and Explosives." When AASTP-1 requirements cannot be met, an ESMRM RSOI Risk Assessment shall be conducted and a risk decision made by the appropriate NATO commander. Chapter 7 provides the requirements for the ESMRM RSOI Risk Assessment. For the purposes of ALP-16:

- a. Munitions reception locations identified during the NATO operational-level planning part of the operational planning process should require that an ESMRM RSOI Risk Assessment be accomplished prior to operation start. Both the risk assessment (to include any risk mitigating actions) and the NATO commander's risk decision can be accomplished before munitions arrive at the LOC node.
- b. Munitions risks identified during RSOI at any LOC node shall be assessed as part of the ESMRM RSOI Risk Assessment.
- c. After munitions have departed the reception or staging area at the LOC point of debarkation, any delays occurring during transportation shall be assessed as part of the ESMRM Transportation Risk Assessment, described later in this chapter.

4.3.3. Risk Acceptance Communication. The ESMRM RSOI Risk Assessment and NATO commander's risk decision shall be provided to the receiving nation's representative responsible for reception (e.g., port authority, officer-in-charge, etc.).

4.4. STORAGE

4.4.1. Munitions Storage. Munitions storage involves the temporary or long-term holding at a designated area in the open or in a structure.

4.4.2. ESMRM Storage Risk Assessment. When NATO explosives safety requirements of AASTP-1 or AASTP-5², as applicable, cannot be met in storage, an ESMRM Storage Risk Assessment shall be conducted and a risk decision made by the appropriate NATO commander. Whenever changes occur that increase the risk (e.g., increased net explosive quantity (NEQ), additional personnel or assets placed

² AASTP-5, "NATO Guidelines for the Storage, Maintenance and Transport of Ammunition on Deployed Missions or Operations." NATO explosives safety criteria found in AASTP-1 and AASTP-5 are based on munitions hazard classified in accordance with the UN hazard classification system and the specific NATO requirements given in AASTP-3.

within AASTP-1 quantity distance (QD) arcs or AASTP-5 field distance (FD) arcs), the assessment shall be updated and reapproved by the appropriate NATO commander. Chapter 7 provides the requirements for the ESMRM Storage Risk Assessment.

4.4.3. Munitions Hazard Classification. All munitions stored as part of a NATO operation shall be hazard classified in accordance with Standardization Agreement (STANAG) 4123, "Determination of the Classification of Military Ammunition and Explosives," and its associated AASTP-3 "Manual of Safety Principles of the Hazard Classification of Military Ammunition and Explosives."

4.4.4. Noncompliance with NATO Hazard Classification Requirements. When NATO operations involve nations that do not use or comply with the UN hazard classification system, STANAG 4123, and AASTP-3, or that do not have adequate surveillance programs (see paragraph 411), it shall be necessary in the ESMRM process to consider the application of more stringent requirements from such nations due to unknown characteristics of their munitions (e.g., HD, storage compatibility group, design, safety).

4.4.5. AASTP-5 Applicability. AASTP-5 provides the requirements and associated reduced FD for munitions storage in a deployed storage operation where NEQ does not exceed 4,000 kilograms per storage cell (barricaded) or in an aboveground location. A storage location can contain any number of such cells and locations and still take advantage of the reduced FD allowed by AASTP-5. If the cell / location NEQ is exceeded, then AASTP-1 QD will apply.

4.4.6. Safety of Stored Munitions. Participating nations with munitions at a NATO base must certify to the NATO base commander that their munitions are safe to store and of known condition (e.g., included in national propellant stability testing and in-service surveillance programs as covered by STANAG 4675, "Principles and Methodology for Ammunition Surveillance in NATO," and its technical documents (Allied Ordnance Publications (AOP)-62, "In-Service Surveillance of Munitions General Guidance," AOP-63, "In-Service Surveillance of Munitions Sampling and Test Procedures," and AOP-64, "In-Service Surveillance of Munitions Condition Monitoring of Energetic Materials." If the owning nation does not (or cannot) provide the aforementioned certification, the unknown condition munitions typically shall not be stored in the same storage location (e.g., cell, magazine) as known condition munitions from nations that have surveillance and propellant stability test programs. A decision to store unknown condition munitions in the same location with known condition munitions shall only be permitted after an ESMRM Storage Risk Assessment has been performed and the NATO base commander has accepted the risk. AASTP-5 provides specific explosives safety requirements related to unknown condition munitions storage at a NATO base.

4.4.7. Protection from the Elements. Since environmental conditions can have a serious impact on munitions' reliability as well as safety (see subparagraph 4.4.6), munitions shall not be exposed unnecessarily to inclement weather or direct sunlight and should be protected from sand, mud, water, and other potential contaminants. Munitions should, whenever possible, be returned to storage in their original packaging and containers, which are designed to protect munitions from improper handling and exposure to environmental conditions.

4.4.8. Depleted Uranium (DU) Ammunition. Refer to AASTP-1 for detailed storage information for DU ammunition.

4.4.9. Captured Enemy Ammunition (CEA). Storing CEA requires two separate approval requirements for movement as well as the NATO base commander's risk decision for storage. Due to CEA's unknown safety condition, CEA presents an increased risk of unintended functioning, fire, and accidents. If the NATO base commander accepts the risk for storing CEA, these munitions should be separated from NATO nations' munitions by inhabited building distance, but never less than intermagazine distance from NATO nations' munitions. AASTP-5 provides requirements related to storage of CEA on a NATO base. AASTP-5 requirements include:

- a. Certification of the CEA by appropriate, qualified personnel as safe for movement and storage.
- b. Risk acceptance for storage by the NATO base commander. The NATO base commander's risk acceptance shall be based on an ESMRM Storage Risk Assessment.

4.4.10. Munitions Loaded Combat Vehicles. Uploaded combat vehicle storage shall comply with NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable. Refer to subparagraph 4.6.6 for requirements applicable to readiness ammunition in uploaded combat vehicles at the unit level. The decision to store uploaded vehicles at other than the unit level as described in subparagraph 4.6.6 shall not be taken lightly as their close proximity almost guarantees complete loss of most, if not all, equipment if an unintended event occurs. If the storage of uploaded vehicles at other than a unit level will be performed, an ESMRM Storage Risk Assessment shall be conducted and risk accepted by the NATO base commander.

4.4.11. Removing Munitions from Military Support Vehicles. Vehicles (military or commercial) in support of NATO military operations, aircraft, or equipment being stored for repatriation from the area of operations (AOO) or disposed of (e.g., sold or transferred in the AOO) shall be searched for munitions to verify their removal³. While waiting for the unloading, the locations where such vehicles are placed (and being unloaded) shall comply with NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable. If those requirements cannot be met, an ESMRM Storage Risk Assessment shall be conducted and risk accepted by the NATO base commander.

4.4.12. Removing Munitions from Battle Damaged Equipment

- a. Battle damaged equipment shall be inspected before storage to verify the removal of all munitions and any other explosives hazards. In addition to explosives concerns, personnel must also recognize that battle damaged equipment may contain additional hazards such as:

³ AMovP-6 provides requirements, as well as inspection checklists, for preparing these vehicles for airlift out of the AOO.

- (1) Human remains
 - (2) Loaded weapons
 - (3) Radioactive material
 - (4) Toxic chemicals and particulate matter
 - (5) Corrosives
 - (6) Fuels, oils, and lubricants
- b. If munitions and other explosives hazards are identified during the inspection, only qualified personnel shall be used to remove these hazards.
 - c. Properly trained personnel shall conduct the munitions and explosives hazard inspection/removal effort as a deliberate task in-situ, at the event location as soon as the equipment and site are secured and the crew evacuated. When munitions and explosives hazard removal cannot be completed on-scene, a cursory inspection shall be conducted at the event location to confirm equipment transport can be accomplished safely before moving to a secure site. The detailed inspection for munitions and explosives hazards shall be conducted in an area that provides appropriate QD and/or explosion effects mitigation.
 - d. Munitions recovered from vehicles must be segregated, further inspected, and final disposition shall be made. Further, immediate actions shall be taken for the disposal of unsafe recovered munitions.

4.5. TRANSPORTATION⁴

4.5.1. Transportation. Transportation is the means of conveyance to move personnel, equipment, and/or stocks (to include munitions), and the requisite materiel handling equipment (from ALP-4.2, paragraph 0605).

4.5.2. ESMRM Transportation Risk Assessment. When the NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met in transportation, an ESMRM Transportation Risk Assessment shall be conducted and a risk decision made by the appropriate NATO commander. Chapter 7 provides the requirements for the ESMRM Transportation Risk Assessment.

4.5.3. Exception and Applicability. While in movement, munitions transportation conveyances (e.g., ships and barges, aircraft, trains) require no QD, but shall comply with the transportation of dangerous goods regulations as contained in AMovP-6.

⁴ Tactical Exception. ESMRM requirements are not applicable to munitions in the tactical transportation phase (e.g., forward arming and refueling point (FARP) or basic load ammunition holding area (BLAHA) moving to a different location). NATO commanders should take actions to minimize munitions risks during tactical situations.

4.5.4. Delays in Munitions Transportation. In the event of unplanned delays during munitions movements (e.g., safe haven, equipment breakdown, accident, emergency), which are expected to exceed 18 hours, an ESMRM Transportation Risk Assessment must be immediately conducted by the responsible authority. Chapter 7 provides required content for an ESMRM Transportation Risk Assessment.

4.5.5. CEA in Transport. CEA transported out of the AOO (e.g., analysis, exploitation) shall comply with movement requirements of AMovP-6⁵, and national regulations, as applicable, shall be followed with regard to minimum markings and symbols, to include the UN hazard classification label, that are required to be applied to shipping packages. Such markings and symbols provide information on the hazards presented by the packaged contents, as well as the storage compatibility group of the munitions concerned when co-located with other dangerous goods. Assignment of the appropriate transportation HD and storage compatibility group is a national responsibility. AASTP-5 provides requirements related to transporting CEA.

4.5.6. Removing Munitions from Military and Military Support Vehicles. Military or commercial vehicles in support of NATO military operations, aircraft, or equipment being repatriated from the AOO or disposed of (e.g., sold or transferred in the AOO) shall be searched for munitions to verify their removal. AMovP-6 provides requirements, as well as inspection checklists, for preparing these vehicles for airlift out of the AOO.

4.5.7. Removing Munitions from Battle Damaged Equipment. Prior to transport of battle damaged equipment, it shall be inspected to verify the removal of all munitions and any other explosives hazards from the equipment. While waiting for the unloading, the locations where such vehicles are placed (and being unloaded) shall comply with NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable. If those requirements cannot be met, an ESMRM Storage Risk Assessment shall be conducted and risk accepted by the NATO base commander. Refer to subparagraph 4.4.12 for additional requirements related to battle damaged equipment.

4.6. DISTRIBUTION

4.6.1. Munitions Distribution. Distribution is the process whereby munitions are obtained from supporting supply points by a unit and broken down for further distribution to subordinate units. Distribution points usually carry no stocks; items drawn are issued completely as soon as possible⁶.

4.6.2. Applicability. An explosives safety site approval or an ESMRM assessment and risk decision is required for munitions distribution locations at an established storage location. Chapter 2 provides additional information related to explosives safety site approvals, which are a national responsibility, utilizing their own site approval processes (Figure 2-3).

⁵ STANAG 2953, "The Identification of Ammunition," and its AOP-2, "The Identification of Ammunition."

⁶ (Allied Administrative Publication (AAP)-6, "NATO Glossary of Terms and Definitions (English & French)," page 2-D-8)

4.6.3. **Unplanned delays** during munitions distribution (e.g., equipment breakdown, accident, emergency) expected to exceed 18 hours require an ESMRM Distribution Risk Assessment. Chapter 7 provides required content for an ESMRM Distribution Risk Assessment.

4.6.4. **ESMRM Distribution Risk Assessment.** When NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met during the distribution phase, an ESMRM Distribution Risk Assessment shall be conducted and a risk decision made by the appropriate NATO commander. Chapter 7 provides the requirements for the ESMRM Distribution Risk Assessment.

4.6.5. **Exception.** An ESMRM risk assessment is not required for the distribution of munitions in a tactical environment where no munitions are stored.

4.6.6. **Readiness Munitions**

- a. Combat units typically hold their readiness basic load ammunition in shipping containers, armored vehicles, trucks, trailers, and structures, or on combat aircraft or re-arming pads. AASTP-1 and AASTP-5 provides storage arrangements and specific QD / FD directly applicable to readiness ammunition locations and Basic Load Ammunition Holding Area (BLAHA).
- b. NATO explosives safety requirements will not generally apply to combat positions (e.g., artillery or mortar firing positions). However, when such locations are on a NATO base, including forward operating bases, NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, apply. If those requirements cannot be met, an ESMRM Distribution Risk Assessment shall be conducted and a risk decision made by the appropriate NATO commander. Chapter 7 provides the requirements for the ESMRM Distribution Risk Assessment.
- c. Readiness munitions can be stored in a BLAHA or in uploaded / combat vehicles, combat aircraft, etc. Any combination of BLAHA with combat loaded vehicle/ combat aircraft or multiple BLAHA are called basic load storage areas. An area with only multiple combat loaded aircraft is called a combat aircraft loading / parking area. However, if these locations are on a NATO base, then explosives safety requirements of AASTP-1 or AASTP-5, as applicable, shall be met. If those requirements cannot be met, an ESMRM Distribution Risk Assessment shall be conducted and a risk decision made by the appropriate NATO commander. Chapter 7 provides the requirements for the ESMRM Distribution Risk Assessment.
- d. Combat and munitions cargo aircraft shall be loaded/unloaded and parked only in designated areas that meet explosives safety requirements. AMovP-6 provides requirements applicable to cargo aircraft. Aircraft containing only installed explosives and safety devices such as authorized signals in survival kits, engine starter cartridges, fire extinguisher cartridges, and other such items necessary to flight operations are exempt from explosives safety requirements.

- e. Munitions should be positioned in designated storage areas near the flight line to meet aircraft loading requirements. Such areas shall meet NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, or must have an approved risk assessment and risk decision. Barricading can be used to further reduce separation distances.
- f. A Forward Arming and Refueling Point (FARP) is the temporary arming and refueling point organized, equipped, and deployed by an aviation unit to support tactical operations. A FARP is typically located closer to the AOO than the combat service support area of an aviation unit and provides fuel and munitions for aviation units in combat. A FARP is excluded from the requirements of ALP-16 per subparagraph 1.3.3.b and Footnote 5 of subparagraph 4.5. However, the term FARP has also been used to describe permanent operations on a NATO base and in such cases, the tactical exemption previously described is not applicable. Such locations shall be required to comply with NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable. If those requirements cannot be met, then the appropriate ESMRM risk assessment (e.g., reception, storage, transportation, distribution, maintenance) shall be conducted and a risk decision obtained in accordance with Chapter 2. Chapter 7 provides the requirements for these types of ESMRM risk assessments.

4.7. MUNITIONS MAINTENANCE AND HANDLING

4.7.1. Munitions Maintenance. Maintenance involves all actions taken to retain munitions or restore these assets to specified conditions until the end of their use, including inspection, testing, servicing, modification(s), classification as to serviceability, repair, recovery, rebuilding, reclamation, salvage, packaging, and repackaging and cannibalization⁷.

4.7.2. Munitions Handling. Maintenance typically involves handling and movement that includes any form of localized movement of munitions (as distinct from transportation) either by purely manual means or with the assistance of mechanical aids to and from:

- a. Production and maintenance facilities
- b. Warehouses and storage
- c. Receiving and shipping areas
- d. Combat and other units

4.7.3. ESMRM Maintenance and Handling Risk Assessment. When NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met during the maintenance and handling phase, an ESMRM Maintenance and Handling Risk Assessment shall be conducted and a risk decision made by the

⁷ AAP-06, "NATO Glossary of Terms and Definitions (English & French)," page 2-M-1.

appropriate NATO commander. Chapter 7 provides the requirements for the ESMRM Maintenance and Handling Risk Assessment.

4.7.4. Removing Munitions from Military Support Vehicles in Maintenance.

Qualified munitions personnel shall inspect and search military or commercial vehicles used in support of NATO military operations and that require maintenance. Any munitions that are found will be removed before maintenance begins. Munitions inspections and temporary holding locations shall meet the NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, or have an ESMRM Maintenance and Handling or Storage Risk Assessment. The type of risk assessment will depend on the type of munitions likely to be encountered, location, expected duration, and proximity of the location to other munitions-related processes and exposures.

4.7.5. Battle Damaged Equipment. Qualified munitions personnel shall inspect and search battle damaged equipment before beginning maintenance to verify the removal of all munitions and any other explosives hazards. Refer to subparagraph 4.4.12 for additional requirements associated with battle damaged equipment.

4.8. RETROGRADE AND REMOVAL

4.8.1. Munitions Retrograde and Removal. Retrograde involves returning munitions from the AOO. Munitions may also be removed from the AOO by disposal (e.g., transfer, sale, destruction), in which case NATO explosives safety requirements of AASTP-1 or AASTP-5 and national requirements will apply.

4.8.2. ESMRM Retrograde and Removal Risk Assessment. When NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met during the retrograde and removal phase, an ESMRM Retrograde and Removal Risk Assessment shall be conducted and a risk decision made by the appropriate NATO commander. Chapter 7 provides the requirements for the ESMRM Retrograde and Removal Risk Assessment.

4.8.3. Applicability. Explosives safety requirements are applicable to each phase of the NATO Consumer Logistics Process, as defined previously in this chapter.

4.8.4. Retrograde Specific Issues. Retrograde specific issues for shipment include:

- a. When the munitions original packaging is no longer available, additional packaging may be required. Since a munition's hazard classification is based on its tested packaging configuration, using a different packaging configuration may change the original hazard classification and compatibility group. Munitions packaging other than the original packaging shall be evaluated by national hazard classification representatives to determine whether a new hazard classification is required.
- b. Before shipping retrograde from the AOO, trained and qualified personnel shall inspect and assess the condition and safety of each munition lot to determine that munitions are safe for shipment. Munitions deemed

questionable or unsafe (e.g., suspended lots, missing components, visible damage, leaking) shall not be retrograded and shall be disposed of in the proper manner.

4.8.5. Removing Munitions from Military Support Vehicles. Military or commercial vehicles used in support of NATO military operations planned for retrograde out of the AOO shall be searched for munitions. Munitions inspections and temporary holding locations shall meet the requirements of AASTP-1 or AASTP-5, as applicable, or have an ESMRM Retrograde Risk Assessment. The type of risk assessment will depend on the type of munitions likely to be encountered, location, expected duration, and proximity of the location to other munitions-related processes and exposures.

4.9. DISPOSAL AND DEMILITARIZATION

4.9.1. Disposal. Disposal encompasses the process of redistributing, transferring, donating, selling, or destroying munitions⁸. Disposal may include end-of-life tasks and actions for munitions resulting from demilitarization operations. STANAG 4518, "Safe Disposal of Munitions, Design Principles and Requirements, and Safety Assessment" provides additional information regarding acceptable NATO disposal methods. If destruction is to be used, Part IV of AASTP-1 provides NATO explosives safety requirements for destruction locations.

4.9.2. Demilitarization. Demilitarization is part of the disposal phase of the NATO Consumer Logistics Process. Demilitarization of munitions is the act of removing or otherwise nullifying the military potential of munitions. Demilitarization is a necessary step for military items prior to their release into a non-military setting⁹.

- a. Demilitarization and destruction of munitions must be carried out according to host nation, national, and NATO requirements. Any dangerous and/or unserviceable munitions shall be destroyed as soon as possible since accumulation presents an unnecessary and additional danger.
- b. A demilitarization facility or destruction area will likely be required for disposing of munitions (e.g., intentional detonation and burning) that is in a dangerous and/or unserviceable condition.
- c. When demilitarization requires intentional detonation, mitigation measures (e.g., burial, orientation, sandbags) shall be used to the maximum extent to manage explosion effects and reduce associated QD. NATO explosives safety requirements for destruction locations and mitigation methods are given in Part IV of AASTP-1.

4.9.3. Battle Damaged Equipment. Battle damaged equipment shall be inspected to verify the removal of all munitions and any other explosives hazards from the

⁸ AOP-38, "Glossary of Terms and Definitions Concerning the Safety and Suitability for Service of Munitions, Explosives and Related Products" page A-26.

⁹ AOP-38, page A-21.

equipment before disposal or demilitarization. Refer to subparagraph 4.4.12 for additional requirements associated with battle damaged equipment.

4.9.4. Disposal and Demilitarization Locations. Locations used for disposal and demilitarization operations shall meet NATO explosives safety requirements of Part IV of AASTP-1 as well as national requirements. Destruction operations shall be carried out only at approved locations (sited locations or those approved by risk decision) and by qualified and trained personnel. NATO explosives safety requirements for destruction locations are given in Part IV of AASTP-1.

4.9.5. ESMRM Disposal and Demilitarization Risk Assessment. When explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met during the disposal phase of the NATO Consumer Logistics Process, an ESMRM Disposal and Demilitarization Risk Assessment shall be conducted and a risk decision made by the appropriate NATO commander. Chapter 7 provides the requirements for the ESMRM Disposal and Demilitarization Risk Assessment.

4.9.6. Record Keeping. Documentation shall be kept of all disposal actions to include, at a minimum:

- a. A log containing the following:
 - (1) Event conducted
 - (2) Date
 - (3) Owning organization and responsible Nation
 - (4) Purpose of disposal action
 - (5) Nomenclature and number of items to be destroyed
 - (6) Munition lots involved
 - (7) Method of disposal used (e.g., burning, detonation, rocket motor firing)
- b. Copies of disposal certification and other associated paperwork.
- c. Disposal reports.

4.10. ESMRM DURING CONCLUSION

4.10.1. Historically, nations involved in operations have not adequately addressed ESMRM requirements during the conclusion of a military operation. Failure to conduct an orderly end to the operation will endanger forces, national civilian populations, and the environment and result in loss of money, equipment, morale, public support, and goodwill.

4.10.2. Paragraph 3.6 of Chapter 3 addresses planning aspects for concluding a military operation.

4.10.3. Chapter 6 provides ESMRM requirements for concluding a military operation.

4.11. APPLICABILITY TO MUNITIONS-RELATED CONTRACT SUPPORT

The NATO explosives safety and ESMRM requirements of this chapter and ALP-16 (subparagraph 1.3.2.c) also apply to munitions-related contract support across all phases of NATO operations and the consumer logistics process in the AOO to include conclusion and post-operation activities. Contractor operations are not exempt from those requirements. Nations and NATO organizations that employ contract support for munitions operations in support of NATO military operations are responsible for contractor compliance with NATO explosives safety, ESMRM, and national requirements.

**CHAPTER 5 EXPLOSIVES SAFETY AND MUNITIONS RISK MANAGEMENT
(ESMRM) PERSONNEL**

5.1. PURPOSE

The purpose of this chapter is to describe the role that ESMRM personnel play in the NATO Operational Planning Process and across the full range of NATO military operations.

5.2. GENERAL

5.2.1. NATO operational missions often involve the use of munitions. Munitions and munition-related processes present a significant, inherent risk and potential consequences include loss of personnel and assets, mission degradation, and political implications. NATO ESMRM policy (Chapter 1) requires that munitions used during NATO operations comply with NATO explosives safety requirements¹. When those requirements cannot be met, the conduct of a Risk Assessment is required to identify munitions-related risks in support of the NATO commander's risk decision.

5.2.2 Qualified ESMRM personnel can:

- a. Identify NATO explosives safety requirements that are applicable to the munitions-related process involved.
- b. Assess if those requirements are being met.
- c. Assist in the conduct of Risk Assessments.

5.2.3 ESMRM reaches across many disciplines and functional areas. Effectively integrating ESMRM throughout NATO requires qualified ESMRM personnel to interface with other communities (e.g., security, intelligence, force protection, logistics, facilities management, military engineers, explosive ordnance disposal (EOD)) to coordinate efforts and communicate munitions-related risks and understand other functional areas requirements and equities.

5.3. MATCHING ESMRM PERSONNEL TO THE TASK

5.3.1. Careful consideration should be given to match ESMRM personnel to the task being conducted. The phase of an operation (e.g., planning, assessment, execution), as well as the type of mission being considered or conducted (e.g., air, sea, ground,

¹ NATO explosives safety requirements are contained in Allied Ammunition Storage and Transport Publication (AASTP)-1, "Manual of NATO Safety Principles for Storage of Military Ammunition and Explosives," AASTP-3, "Manual of NATO Safety Principles for the Hazard Classification of Military Ammunition and Explosives," AASTP-4, "Manual on the NATO Principles for the Application of Risk Analysis to the Storage and Transportation of Military Ammunition and Explosives," and AASTP-5, "NATO Guidelines for the Storage, Maintenance and Transport of Ammunition on Deployed Missions or Operations."

transportation), will determine the ESMRM personnel skill sets needed to address explosives safety aspects associated with that task.

5.3.2. Paragraph 5.5 provides examples of some explosives safety / munition-related personnel skill sets that exist within NATO nations to support ESMRM functions. Each personnel skill type provides unique capabilities; consequently, it should never be assumed that an individual assigned to one role has the required qualifications to perform another. The qualifications of individuals to perform assigned tasks must be assessed on a role-by-role basis.

5.3.3. When considering the ESMRM personnel skills needed, relevant aspects of the phase / mission shall be considered (e.g., disposal, airbase operations, port operations, transportation).

5.4. QUALIFIED PERSONNEL

5.4.1. Nations that transport, maintain, handle, store, use, demilitarize and dispose of munitions understand the need to train their own workforce to accomplish specific munitions and explosives safety-related tasks effectively, efficiently and safely. Nations assure professional competence by developing and providing a training curriculum specific to the individual task or by sending their munitions-related personnel to attend another nation's munitions training courses.

5.4.2. Nations have a responsibility (subparagraphs 2.6.2.a(1)(a) and 2.7.2.a(2)) in NATO to provide qualified personnel to perform ESMRM tasks. In the absence of qualified personnel, formal arrangements (e.g., memorandums of understanding / agreement) shall be established with other munitions-contributing nations to provide qualified personnel to manage and assure compliance with NATO explosives safety requirements.

5.5. ESMRM-RELATED PERSONNEL

The following are descriptions of some of the personnel skill sets may be available to assist with ESMRM functions. For munitions-related personnel descriptions or skills not listed below, personnel training and qualifications must be reviewed by national appointing authorities to assess an individual's ability to perform specific ESMRM functions.

- a. Explosives Safety Officer (ESO). An ESO is a highly trained civilian or military individual designated in writing to advise a commander on explosives safety matters. The ESO provides explosives safety management and oversight of the commander's ESMRM program. Minimum competencies required by an ESO are detailed at Annex B of AASTP-5². An individual qualified to be an ESO would have the skills necessary to support ESMRM throughout all phases. At a NATO base with two or more nations present and where munitions are located and munitions-related processes occur (subparagraph 2.6.2), the NATO base

² Allied Ammunition Storage and Transport Publication AASTP-5, "NATO Guidelines for the Storage, Maintenance and Transport of Ammunition on Deployed Missions or Operations."

Commander will appoint or direct the appointment of a Lead ESO to execute the NATO base's ESMRM program.

- b. Ammunition Technical Officer (ATO). An ATO is generally a military officer who has received formal munitions training. The level of training may vary by nation but generally, a fully trained ATO is well-qualified to serve as an ESO and brings many other qualification and skills that can enhance the ESMRM process. Minimum competencies for an ESO as described above in subparagraph 5.5.a must be met.
- c. Service-trained Ammunition-related Military Personnel. Many nations provide Service-specific munitions training for their officers/noncommissioned officers relating to the functions they serve (e.g., ground, air, sea). Qualifying service-trained individuals as an ESO requires comparing an individual's training against the ESO competencies described above in subparagraph 5.5.a.
- d. Ammunition Technician (AT). An AT generally describes a military or civilian an individual within the munitions supply chain who works at a fixed location such as an ammunition supply point or ammunition storage depot. ATs are capable of providing technical expertise at the installation (e.g., logistics disposal, inventory control, receipts and issues, maintenance). The level and scope of training will vary by nation. An assessment of an individual's training, skills, and experiences must be made against ESO minimum competencies however, a fully trained and experienced AT should qualify to serve as an ESO as described above in subparagraph 5.5.a.
- e. Safety Engineer/Technicians/Quality Assurance Specialist. Highly trained individuals who provide explosives safety support to deployed forces from home locations. They may have also deployed to provide in-theater explosives safety support to their national forces. Safety Engineer/Technicians/Quality Assurance Specialist could be identified as ESOs, provided their training and experience meet minimum competencies for an ESO as described above in subparagraph 5.5.a.
- f. Ammunition Handler. Ammunition handlers handle munitions and typically receive basic explosives safety hazards training, warehousing and firefighting skills, and learn how to operate material handling equipment. An ammunition handler would not have the necessary skills to be an ESO as described above in subparagraph 5.5.a.
- g. EOD. EOD personnel, although highly trained for the destruction and disposal of ammunition, would not generally qualify as ESOs as described above in subparagraph 5.5.a, unless they had previously received ATO and/or AT training. EOD personnel in-depth understanding of munitions, the hazardous effects produced, and mitigation/protection of those effects can make them invaluable members of any ESMRM team. EOD personnel shall be consulted when the destruction of munitions (e.g. intentional munitions burn or detonation) is part of an ESMRM assessment.

- h. Contractors. Contractors can play a significant role in ESMRM efforts and may provide a service that is not readily otherwise available to a nation. Typically, contracted individuals would not be assigned as ESOs. However, if a contracted individual has the necessary experience and qualifications to meet the competencies for an ESO as described above in subparagraph 5.5.a, then that individual can be assigned to fill an ESO position.

CHAPTER 6 INTEGRATING EXPLOSIVES SAFETY AND MUNITIONS RISK MANAGEMENT (ESMRM) REQUIREMENTS INTO THE CONCLUSION OF A NATO MILITARY OPERATION

6.1. PURPOSE

This chapter provides NATO ESMRM requirements for munitions and munitions-related processes and activities during the conclusion of a NATO military operation at forward operating bases (FOBs) and forward logistics bases (FLBs) as part of the redeployment and disestablishment stage of a NATO military operation. Chapter 6 also clarifies that munitions processes and activities performed by other than military personnel under contract must also comply with the requirements contained herein.

6.2. ESMRM REQUIREMENTS DURING CONCLUSION

6.2.1. As stated in Chapter 3, preparing for the conclusion of munitions-related activities at operational locations must be planned for during the initial analysis of the desired end state, together with the means to achieve it. Consideration of the effort required cannot be overstated. Concluding a military operation can be as complicated as deployment and is often not adequately planned for in advance.

6.2.2. ESMRM requirements included in this chapter also apply to the closure or changes associated with any NATO base regardless of whether or not an operation is ongoing.

6.2.3. Figure 6-1 illustrates the relationship between the NATO Operational Stages, the NATO Consumer Logistics Process phases, and ESMRM applicability across the full range of NATO military operations. The circle further emphasizes additional requirements for the redeployment and disestablishment operational stages contained in Chapter 6.



Figure 6-1. Relationship of ESMRM Process to the NATO Operational Stages and the Consumer Logistics Process

6.2.4. At the conclusion of operations, NATO commanders of FOBs and FLBs shall implement the Munitions Retrograde Plan developed during the operational planning process (OPP) and terminate ongoing munitions-related consumer logistics processes identified in Chapter 4 and Figure 6-1.

6.3. Planning for Conclusion

6.3.1. Munitions retrograde shall be planned during the OPP and a Munitions Retrograde Plan shall be developed. Munitions Retrograde shall be addressed in the ESMRM section or annex for all NATO plans where NATO munitions use is anticipated.

6.3.2. Chapter 3 provides detailed requirements for the Munitions Retrograde Plan.

6.3.3. At the conclusion of NATO military operations (i.e., redeployment and disestablishment portion of the NATO Operational Stages), the reverse of the NATO Consumer Logistics Process that delivered supplies (to include Class V munitions) to the operational theatre occurs.

6.3.4. The following munitions and munitions-related processes and activities associated with the redeployment and disestablishment stages of concluding a military operation are linked to the Consumer Logistics Process:

- a. Consolidation and Collection (Reception) involving the relocation of munitions to central locations for further action.
- b. Storage of munitions in preparation for retrograde or disposal (e.g., transfer, sale, demilitarization, destruction). Storage is likely to be temporary in nature and care must be exercised since munitions turned in will be done by a large number of units.
- c. Transportation associated with redeployment and disestablishment stages.
- d. Distribution of munitions back to consolidation, collection, and storage locations as part of unit turn-ins.
- e. Maintenance of munitions to include packaging and preparation by units prior to turn-in.
- f. Retrograde of munitions by units through lines of communication (LOC) to the country of origin or final destination.
- g. Disposal of munitions (e.g., transfer, sale, demilitarization, and destruction).

6.4. CONSOLIDATION AND COLLECTION (RECEPTION)

6.4.1. For the purposes of ALP-16, consolidation and collection is the equivalent of reception during the redeployment and disestablishment stages at the conclusion of a NATO military operation.

6.4.2. Consolidation and collection locations must meet NATO explosives safety requirements of Allied Ammunition Storage and Transport Publication AASTP-1¹ or AASTP-5², as applicable, or have an ESMRM RSOI Risk Assessment and risk decision by the NATO commander. Chapter 7 provides the requirements for the ESMRM RSOI Risk Assessment.

6.4.3. ESMRM risk assessments are an integral part of managing the potential risks during the consolidation and collection phase and may need to be performed for these locations. Actions typically associated with consolidation and collection include:

- a. Reception of all munitions at munitions storage areas. Munitions not returned in their original sealed packaging shall be assumed to be in an “unknown” condition.
- b. Inspection of munitions to determine condition. During inspection of returned munitions, a qualified technical expert shall assess the condition of returned munitions with special regard to it being safe to move on the imminent retrograde movement.
- c. Repacking of serviceable munitions. Munitions requiring repackaging must be packaged into containers that meet United Nations and International Organization for Standardization requirements for storage and transport.
- d. Arrangements for any required disposal (e.g., transfer, sale, demilitarization, destruction).

6.5. STORAGE

6.5.1. Storage locations must meet NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, or have an ESMRM Storage Risk Assessment and risk decision by the NATO commander. Chapter 7 provides the requirements for the ESMRM Storage Risk Assessment.

6.5.2. Storage of munitions is the same throughout the deployment, redeployment, and disestablishment stages of a NATO military operation. Storage during the redeployment and disestablishment stage is likely to be temporary and dynamic due to unforeseen munitions turn-ins. Areas not previously planned for munitions storage

¹ AASTP-1 “Manual of NATO Safety Principles for Storage of Military Ammunition and Explosives”

² AASTP-5, “NATO Guidelines for the Storage, Maintenance and Transport of Ammunition on Deployed Missions or Operations”

may need to be used to accommodate higher volumes of munitions storage. Assessments of munitions storage locations shall also focus on the following prior to redeployment:

- a. Condition of munitions
- b. Condition of packaging
- c. Surveillance information
- d. Hazard classification
- e. Expected munitions throughput. During retrograde, significant quantities of munitions may be delivered to a storage location at a rate far beyond the capabilities of that location. Increased throughput will likely cause licensed explosives limits to be exceeded, resulting in violations of established quantity-distance and may exacerbate or worsen safety cases. Increasing throughput will likely require a new ESMRM Storage Risk Assessment.

6.5.3. Previously approved ESMRM risk assessments and risk decisions will likely need to be updated during the redeployment and disestablishment stage due to the dynamically changing operational environment.

6.5.4. Refer to AASTP-1 for detailed storage information for depleted uranium munitions stored during the redeployment and disestablishment stage.

6.6. TRANSPORTATION

6.6.1. Although transportation requirements of munitions are the same throughout the deployment, redeployment, and disestablishment stages of a NATO military operation, there may be additional ESMRM considerations during the redeployment and disestablishment stages that could require modifications of existing ESMRM assessments or performance of new assessments. Examples of modifications include but are not limited to:

- a. Hazard classification if repackaged in other than original packaging. A munition's assigned hazard classification is only valid in its tested, packaged configuration. If not in that configuration, the hazard classification is no longer valid and the munitions may actually present a very different threat (e.g., hazard division (HD) 1.3 or storage sub-division (SsD) 1.2.1 may now be HD 1.1). See AASTP-3³, for additional details.
- b. Foreign munitions and weapons systems requiring intelligence exploitation.

6.6.2. When required munitions locations associated with transportation (e.g., safe haven, unforeseen delays, temporary holding) during the redeployment and

³ AASTP-3, "Manual of NATO Safety Principles for the Hazard Classification of Military Ammunition and Explosives"

disestablishment stage must meet NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, or have an ESMRM Transportation Risk Assessment and risk decision by the NATO commander. Chapter 7 provides the requirements for the ESMRM Transportation Risk Assessment.

6.6.3. Movement of munitions during the redeployment and disestablishment stage shall comply with transportation of dangerous goods regulations as contained in Allied Movement Publication (AMovP)-6⁴.

6.7. DISTRIBUTION / COLLECTION

6.7.1. During the redeployment and disestablishment stage, distribution is not a key phase of the Consumer Logistics Process, rather, collecting munitions becomes the primary function. During the Distribution / Collection stage, the focus is on units returning munitions (e.g., excess, unaccounted for, or other types of munitions) to consolidation and storage locations in preparation for transportation.

6.7.2. Distribution / Collection locations must meet NATO explosives safety requirements of AASTP-1 or AASTP-5, as appropriate, or have an ESMRM Distribution/Collection Risk Assessment and risk decision by the NATO commander. Chapter 7 provides the requirements for the ESMRM Distribution / Collection Risk Assessment.

6.7.3. A major challenge during the redeployment and disestablishment stage at the unit level is munitions returned in unknown condition, and in varying configurations (e.g., unpackaged, improper packaging, flares without safety caps, taped grenades). To preclude hazardous conditions during the redeployment and disestablishment stage, munitions personnel must make every effort throughout the operational stage to maintain original munitions packaging materials.

6.7.4. The Munitions Retrograde Plan shall address collection of munitions during the redeployment and disestablishment stage.

6.8. MAINTENANCE

6.8.1. Maintenance locations must meet NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, or have an ESMRM Maintenance Risk Assessment and risk decision by the NATO commander.

6.8.2. During the redeployment and disestablishment stage, maintenance of munitions will shift from preparing munitions for employment to inspecting, repackaging, and unit return in preparation for demilitarization, sale, transfer, or retrograde. Munitions maintenance during the redeployment and disestablishment stage may be done at a separate munitions operating location or as part of a storage operation.

⁴ AMOV P-6, Allied Movement Publication 6, "Allied Multimodal Transportation of Dangerous Goods Directive."

6.8.3. A separate ESMRM assessment may need to be developed or the existing ESMRM Maintenance Risk Assessment may need to be modified during the maintenance stage.

6.9. RETROGRADE

6.9.1. ESMRM personnel will determine whether existing risk assessments (e.g., transportation, storage, distribution, and maintenance) and risk decisions need updating. If existing ESMRM risk assessments require updating, ESMRM personnel shall take actions to do so. ESMRM personnel shall also conduct new ESMRM assessments and obtain NATO commander risk decisions, as necessary.

6.9.2. Munitions may also be removed from the area of operations (AOO) in which case NATO explosives safety requirements of AASTP-1, AASTP-3, AASTP-5, ESMRM, and national requirements also apply.

6.9.3. Retrograde during the redeployment and disestablishment stage involves returning munitions from the AOO. During the redeployment and disestablishment stage, retrograde involves transporting munitions from LOC nodes (e.g., airport of debarkation, seaport of debarkation, railhead) to their final destination.

6.10. DISPOSAL

6.10.1. **Complying with NATO or National Requirements.** During the redeployment and disestablishment stage, disposal of munitions (e.g., transfer, sale, demilitarization, and destruction) within the AOO becomes a major function. See AOP-38⁵. Standardization Agreement 4518, "Safe Disposal of Munitions, Design Principles and Requirements, and Safety Assessment" provides information related to demilitarization that does not involve destruction. Part IV of AASTP-1 provides specific NATO explosives safety requirements for destruction areas (i.e., open burning (OB) and open detonation (OD)).

- a. When arranging for disposal (e.g., transfer, sale, demilitarization, destruction), nations shall comply with NATO explosives safety requirements of AASTP-1 or AASTP-5, as appropriate, and other existing requirements (e.g., environmental).
- b. Some nations' regulations do not allow destruction of surplus munitions by OB or OD. In this case, munitions must meet transportation requirements and risk assessments/decisions, as applicable. Safety must always be paramount so that dangerous munitions are not transported, with the possibility of creating potentially damaging situations.

6.10.2. **ESMRM Disposal and Demilitarization Risk Assessment.** When NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be

⁵ Allied Ordnance Publication-38, "Glossary of Terms and Definitions Concerning the Safety and Suitability for Service of Munitions, Explosives and Related Products."

met during the retrograde and removal phase, an ESMRM Disposal and Demilitarization Risk Assessment shall be conducted and a risk decision shall be made by the appropriate NATO commander. Chapter 7 provides the regulations and requirements for the ESMRM Disposal and Demilitarization Risk Assessment.

6.10.3. **Physical Destruction.** In the event munitions require physical destruction through OB or OD, the location chosen for destruction must meet the requirements of AASTP-1 Part IV, or have an ESMRM Disposal and Demilitarization Risk Assessment and risk decision by the NATO commander.

6.11. POST-OPERATION ACTIVITIES

6.11.1. **Executing the Munitions Retrograde Plan during Post-Operation Activities.** In the event of post-operation activities, the Munitions Retrograde Plan shall be used to address ESMRM aspects associated with any remaining munitions-related processes and activities. Examples include:

- a. Infrastructure. The potential transfer or destruction of any munitions operations infrastructure built by NATO member nations in support of the mission. Any residual munitions-related risks should be removed or communicated and accepted by the receiving nation.
- b. Removal of explosive remnants of war. Any residual munitions-related risks from explosive remnants of war should be removed or communicated and accepted by the receiving nation.
- c. Closing or transferring training ranges. Any residual munitions-related risks from ranges where munitions were used should be removed or communicated and accepted by the receiving nation.
- d. Any additional ESMRM issues not previously identified/addressed.

6.11.2. **Continuing Munitions-Related Processes and Activities During Post-Operation Activities.** For the purposes of ALP-16, post-operation activities involving munitions, munitions-related processes or activities fall into one of the Consumer Logistics Process phases described in Chapter 6 and Chapter 4.

- a. In the event that munitions-related activities are present during post-operation activities involving munitions, these processes must meet the NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, or have the appropriate ESMRM risk assessment and risk decision by the NATO commander.
- b. Post-operation activities may include:
 - (1) Belligerent occupation
 - (2) Humanitarian relief
 - (3) Civil administration

- (4) Demobilization operations
- (5) Battle area clearance

CHAPTER 7 THE NATO EXPLOSIVES SAFETY AND MUNITIONS RISK MANAGEMENT (ESMRM) RISK ASSESSMENT PROCESS

7.1. PURPOSE

The purpose of this chapter is to describe and define the process for conducting NATO planning and operational ESMRM risk assessments when NATO explosives safety requirements of Allied Ammunition Storage and Transport Publication (AASTP-1)¹ and AASTP-5² cannot be met in preparation for the NATO commander's risk decision. The process consists of discrete steps that incorporate the risk analysis methodology described in AASTP-5.

7.2. GENERAL

7.2.1. NATO Explosives Safety Requirements. In the planning and execution of NATO military operations, situations are likely to exist when NATO explosives safety requirements cannot be met. In such cases, risks to and from munitions-related operations must be assessed in support of the NATO commander's risk decisions.

7.2.2. Munitions Storage. When NATO nations with co-located munitions storage and storage-related processes determine that explosives safety requirements of AASTP-1³ or AASTP-5 cannot be met, the ESMRM process defined and described in this chapter shall be used to assess and document the risk, potential consequences, decision-making, and any associated mitigation and related controls.

7.2.3. Lines of Communication (LOC). Assessing the munitions and munitions-related process of transportation at strategic LOC provides the NATO commander with an overview of LOC as a single system. The ESMRM risk assessment process includes important information such as vulnerabilities, hazards to operations, exposures to unrelated personnel, and infrastructure in relation to the NATO mission. Typically, LOC include but are not limited to:

- a. Seaports of embarkation/debarkation
- b. Airports of embarkation/debarkation
- c. Railheads
- d. Roads
- e. Temporary holding, staging, and transfer areas

¹ Allied Ammunition Storage and Transport Publication AASTP-1, "Manual of NATO Safety Principles for Storage of Military Ammunition and Explosives"

² AASTP-5, "NATO Guidelines for the Storage, Maintenance and Transport of Ammunition on Deployed Missions or Operations"

7.2.4. Other Munitions-Related LOC Activities. Although munitions storage and movement through LOC are the most likely activities to require application of the ESMRM risk assessment process, there are other life-cycle activities associated with LOC that may require the risk assessment process be applied to them including maintenance and handling, retrograde and removal, and disposal (e.g., transfer, sale, demilitarization, destruction).

7.2.5. Risk-Decision Making. The ESMRM risk assessment process detailed and described in this chapter has been developed to provide NATO commanders and national leaders with a consistent and documented methodology in support of their risk decisions. The process uses a variety of quantitative and qualitative tools, to include AASTP-1 quantity-distance (QD) and AASTP-5 field distance (FD), and risk-based tools, observations, interviews, and information gathering before and during the assessment, as well as analysis and application of NATO explosives safety requirements in AASTP-1 and AASTP-5.

7.2.6. ESMRM Roles, Responsibilities, and Requirements. Chapter 2 provides the NATO operational chain-of-command ESMRM roles, responsibilities, and requirements.

7.3. ESMRM RISK ASSESSMENT CATEGORIES AND TYPES

7.3.1. Planning Risk Assessment. The Planning Risk Assessment is the first of two categories of possible ESMRM risk assessments and is part of the operational or contingency planning process. The purpose of the Planning Risk Assessment is to identify potential risks from planned munitions-related activities (e.g., munitions transport through an LOC, establishment of a munitions storage area at a planned NATO base). There are three types of ESMRM Planning Risk Assessments:

- a. Strategic Operations Planning
- b. Operational-Level Planning
- c. Tactical Operations Planning

7.3.2. Operational Risk Assessment. The Operational Risk Assessment is the second category of possible ESMRM risk assessments. The Operational Risk Assessment is conducted after the start of the mission in the area of operation (AOO) to assess existing or planned munitions-related operations that are known not to meet NATO explosives safety QD criteria. The purpose of the Operational Risk Assessment is to identify potential risks from existing or planned munitions-related activities after the start of an operation. There are seven types of ESMRM Operational Risk Assessments:

- a. Reception, staging, onward movement and integration (RSOI)
- b. Storage
- c. Transportation

- d. Distribution (Note: During the redeployment and disestablishment stage, distribution is not a key phase of the Consumer Logistics Process, rather, collecting munitions becomes the primary function. For that reason, a risk assessment conducted for this stage is called Distribution / Collection Risk Assessment)
- e. Maintenance and Handling
- f. Retrograde and Removal
- g. Disposal and Demilitarization

7.4. RISK ASSESSMENT MAINTENANCE AND UPDATE FREQUENCY

Strategic, operational, and tactical environments are dynamic and fluid; therefore, ESMRM risk assessments shall be continuously maintained and updated since the supporting information is likely to change over time resulting in outdated recommendations. A reevaluation of an approved ESMRM risk assessment and its decision shall be performed to determine if an update or new risk assessment / risk decision is required when one of the following occurs:

- a. Twelve months have passed since the last risk assessment. ESMRM risk assessments shall be routinely reevaluated every 12 months.
- b. Change of NATO commander or delegated risk decision authority.
- c. Change in operational and / or contingency plans.
- d. Changes at a specific location occur that affect personnel, equipment, or infrastructure and potentially increase the risk and consequences from an unintended explosives event. Change could be mission driven (e.g., change in LOC location, increased net explosive quantity (NEQ), presence of more hazardous munitions, increased personnel and equipment exposed) or externally driven (e.g., increased public exposure).

7.5. RISK ASSESSMENT PROCESS

A successful ESMRM risk assessment requires performing 12 discrete steps as shown in Figure 7-1 and described in this chapter.

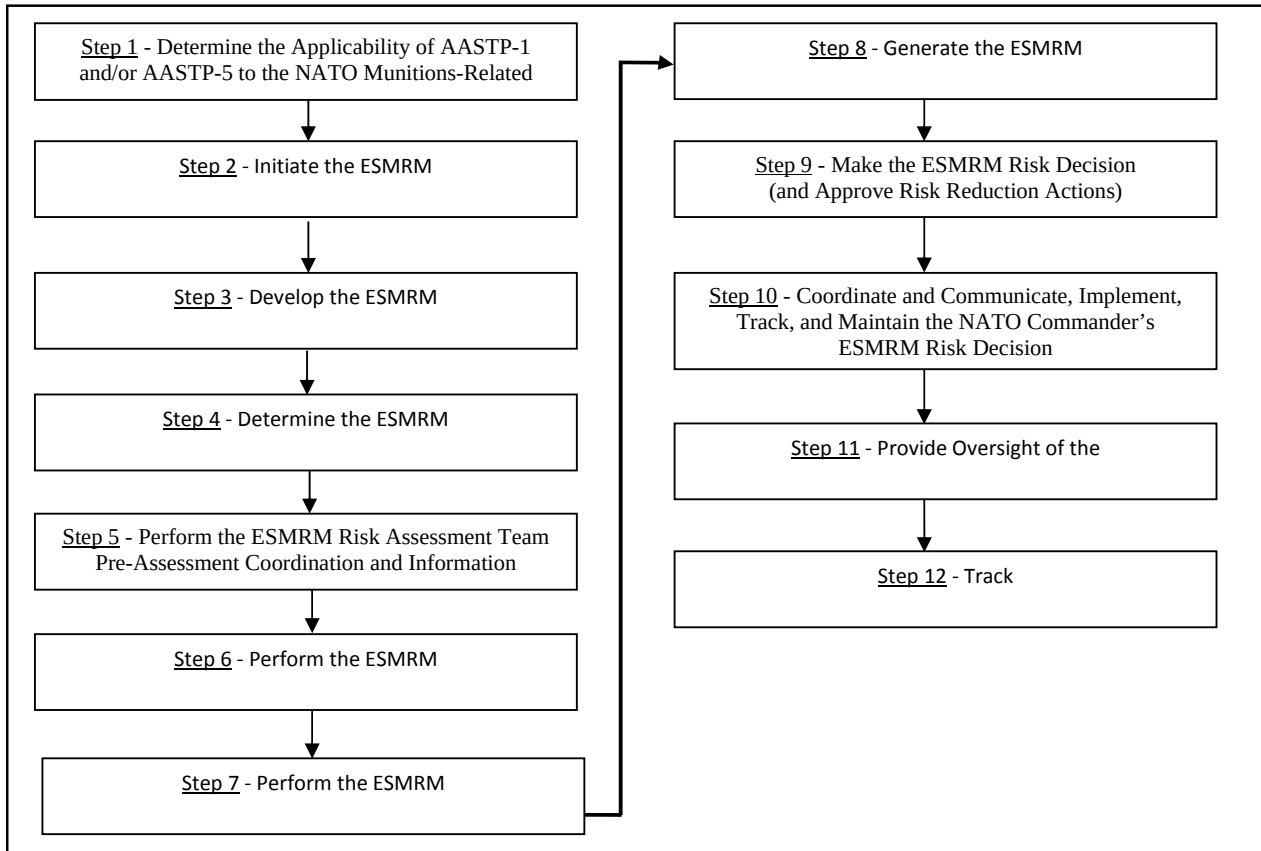


FIGURE 7-1. ESMRM RISK ASSESSMENT PROCESS FLOW

7.6. STEP 1. DETERMINE THE APPLICABILITY OF AASTP-1 OR AASTP-5 TO THE NATO MUNITIONS-RELATED PROCESS

The NATO commander in coordination with his staff shall:

- a. Determine whether NATO explosives safety requirements of AASTP-1 or AASTP-5 apply to munitions-related processes.
- b. Determine if the applicable requirements of AASTP-1 or AASTP-5 can be met for munitions-related processes.
 - (1) If NATO explosives safety requirements can be met, the owning nation shall follow national requirements to license munitions storage and operations per AASTP-1 or AASTP-5, as applicable.
 - (2) If NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable, cannot be met, the ESMRM risk assessment process described herein must be performed.

7.7. STEP 2. INITIATE THE ESMRM RISK ASSESSMENT

7.7.1. As stated in Chapter 2, when NATO explosives safety requirements cannot be met the NATO commander shall direct his staff to assess munitions-related risks and generate a risk assessment in preparation for a risk decision.

7.7.2. The ESMRM risk assessment will usually be conducted with available organic staff however, in the event organic staff is not available, the NATO commander may request additional resources through NATO and national chains of command.

7.7.3. In the absence of such support, or in the event specialized expertise of a multinational nature is required, the NATO commander may request assistance from the NATO AC/305 (Logistics Committee (LC)), which can request support from its ESMRM Panel, as well as from NATO AC/326, Conference of NATO Armaments Director Ammunition Safety Group, which produces NATO explosives safety requirements.

7.8. STEP 3. DEVELOP THE ESMRM RISK ASSESSMENT SCOPE

7.8.1. **ESMRM Scope.** The lead will coordinate the development of the risk assessment scope of work and obtain the requesting NATO commander's approval. The scope shall include:

- a. Scope content
- b. Signature page for the NATO commander and the designated lead for the agreed output
- c. Munitions-related processes and locations requiring assessment
- d. Methodology
- e. Assessment team composition
- f. Timelines (e.g., assessment and deliverables)
- g. Deliverables (e.g., report and briefs)
- h. Any required follow-on actions

7.8.2. **Modifications to the ESMRM Scope.** Recognizing the reality and dynamic nature of assessments as well as the conditions at locations identified in the scope, the scope may require modification:

- a. All modifications to the scope shall be documented for complete understanding and become part of the risk assessment report.
- b. Both the requestor and the lead shall agree to all modifications.

7.9. STEP 4. DETERMINE THE ESMRM RISK ASSESSMENT TEAM COMPOSITION

Team composition is likely to vary depending on the category and type of risk assessment conducted. Table 1-1 illustrates the types of ESMRM risk assessments that may be required. The lead will assemble and determine the team composition for the type of ESMRM risk assessment to be conducted. The team shall consist of personnel capable of assessing munitions-related processes and their associated risks. Chapter 5 provides details about ESMRM personnel skill sets available to support ESMRM risk assessments.

- a. Planning Risk Assessment Team. Planning Risk Assessment team members may include, but are not limited to, NATO and national representatives and experts from:
 - (1) Involved commands
 - (2) Explosives Safety, ESMRM qualified personnel
 - (3) Logistics
 - (4) Transportation
 - (5) Military Engineers
 - (6) Security/Force Protection
 - (7) Intelligence
 - (8) Geospatial (for mapping)
 - (9) Legal
 - (10) Protocol
- b. Operational Risk Assessment Team. Since the Operational Risk Assessment involves assessing munitions-related processes at a NATO base (e.g., forward operating base, activity, installation), personnel familiar with the processes shall participate. Team members may include, but are not limited to, NATO and national representatives and experts from:
 - (1) Involved commands
 - (2) Explosives Safety, to include the Lead explosives safety officer (ESO) (see Chapter 2)
 - (3) Logistics
 - (4) Transportation

- (5) Military Engineers
- (6) Security/Force Protection
- (7) Intelligence
- (8) Geospatial (for mapping)
- (9) Legal
- (10) Protocol
- (11) Base operations support, to include the base operations support integrator (BOSI) (see Chapter 2)

7.10. STEP 5. PERFORM THE ESMRM RISK ASSESSMENT TEAM PRE-ASSESSMENT COORDINATION AND INFORMATION GATHERING

Advance arrangements and coordination are needed in preparation for travel to the location and to collect site-related information required to perform the risk assessment. Required information is likely to include, but is not limited to:

a. Coordination with External Organizations

- (1) NATO and national operational elements involved with or supporting the operation
- (2) National explosives safety specialists (coordinated through nations involved with or supporting the operation)
- (3) Military engineer elements
- (4) NATO and national protocol
- (5) NATO and national intelligence elements
- (6) National Geospatial-Intelligence and mapping elements
- (7) Host nation supporting elements

b. Logistics Requirements. Logistics requirements may include, but are not limited to:

- (1) Medical (e.g., vaccinations, certificates)
- (2) Country clearance
- (3) Personal protective equipment
- (4) Transportation and hotel/housing

- (5) Advance notifications
 - (6) Applicable restrictions and limiting factors
 - (7) Host nation and local requirements
 - (8) Political conditions (country brief)
 - (9) Training (e.g., intelligence, security, force protection, NATO-specific)
 - (10) Equipment critical to mission success (e.g., global positioning system, camera, laptop computer, range finder, communications equipment)
 - (11) Personal security clearance information
- c. Pre-Assessment Information Gathering. Pre-assessment information gathering requirements may include, but are not limited to:
- (1) Munitions and munitions-related processes
 - (2) Existing deviations, ESMRM risk assessments, and consequence acceptance documents
 - (3) Operations plan details and supporting information
 - (4) Relevant concept of operations for exercise or other military operation
 - (5) Maps and overhead imagery
 - (6) Supporting infrastructure relating to munitions and munitions processes
 - (7) Status-of-forces and other international agreements
 - (8) Host nation munitions and munitions processes information
 - (9) Local host nation logistic node laws and regulations
 - (10) Host nation explosives safety laws, limitations, and regulations
 - (11) Exposures (e.g., population density, vehicles, infrastructure).
 - (12) Coordination with external organizations.

7.11. STEP 6. PERFORM THE ESMRM RISK ASSESSMENT

7.11.1. Annex A provides the ESMRM Risk Assessment Form for use when performing risk assessments for munitions and munitions-related functions. Annex B provides an example of a completed ESMRM Risk Assessment Form.

7.11.2. Table 1-1 identifies the types of ESMRM risk assessments associated with the NATO Operational Planning Process (OPP), Consumer Logistics Process, and Operational Stages.

7.11.3. Information gathered about the locations during the pre-assessment step shall be reviewed in preparation for the on-site assessment.

7.11.4. Once on-site, the team or its representatives shall collect site-specific technical information that may include, but is not limited to:

- a. RSOI and required equipment (e.g., heavy equipment transports)
- b. Supporting equipment necessary to enable the offload of containerized munitions (e.g., cranes, material handling equipment)
- c. Tactical assembly areas and large gun siting/checkout areas as part of RSOI
- d. Supporting munitions-enabling infrastructure
- e. Roads for munitions transport to include width assessment based on the type of vehicles used
- f. Storage pads/area
- g. Surface transportation routes of ingress/egress (e.g., rail or road)
- h. Availability of emergency response equipment
- i. Location of and information on potential exposed sites such as facilities, hospitals, schools, and houses (see subparagraphs 7.11.5. and 7.11.6. for appropriate assessment distances)
- j. Location of hazardous materials
- k. Commercial operations
- l. Placement of utilities (e.g., gas pipes, power stations, electrical lines)
- m. Clear zones around unloading and loading points.
- n. Location of critical communications equipment.
- o. Impact of munitions and munitions-related processes on the rest of the mission.
- p. Ability to access the unloading and loading points.
- q. Lightning protection systems.

- r. Risks to munitions and munitions-related processes. (e.g., power lines, liquefied natural gas, refineries, grain elevators, bulk fuels facilities, factories).

7.11.5. Exposures within a predetermined distance of munitions-related processes shall be assessed. Predetermined distances shall not be less than the inhabited building distances (IBDs) provided in AASTP-1 or AASTP-5, as appropriate, for the hazard divisions (HDs) and maximum NEQ expected.

7.11.6. When HD 1.1 is involved, the assessment area shall examine distances of two times the IBD (2 x IBD) to determine vulnerable construction as defined in Allied Ordnance Publication 38, "Glossary of Terms and Definitions Concerning the Safety and Suitability for Service of Munitions, Explosives and Related Products," and as further described in AASTP-1.

7.12. STEP 7. PERFORM THE ESMRM RISK ANALYSIS

Determining Risk Level through Analysis. An explosives safety risk analysis is a systematic process that determines the actual level of the hazard for a given situation. It takes into account possible adverse effects (consequences) produced by an unintended explosion or fire as well as how often such effects may occur (frequency/probability). The explosives safety risk analysis process is generally a combination of quantitative calculation where the data and tools are available and qualitative assessment of other relevant information, taking into account factors such as operational requirements.

- a. Paragraph 2.7.2 of AASTP-5 provides information about the qualitative risk analysis approach. Paragraph 2.7.3 of AASTP-5 provides a description of a quantitative risk analysis method for the storage of ammunition on deployed missions⁴.
- b. To determine the appropriate risk level (i.e., high, medium, or low) for the risk assessment, the assessment team shall determine the probability that an unintended event will occur as well as fit the determined risk into the proper consequence category shown in Table 7-1. The intersection of probability and consequence in Table 7-1 provides the risk level. Probability definitions are provided in Table 7-2; consequence definitions are given in Table 7-3.
- c. The probability of an event can be determined in two ways (or a combination of both): by historical record of events or by an analytical

⁴ AASTP-5 also describes/provides two qualitative tools that can be used to determine the consequence portion of assessments to help determine the severity of an incident. However, that method is limited to a maximum NEQ of 4,000 kilograms HD 1.1 per storage cell. If the cell/location NEQ is exceeded, then AASTP-1 QD will apply. When those QDs cannot be met, the only options are qualitative risk assessments as detailed in AASTP-5, or quantitative risk assessments using AASTP-4, "Manual on the NATO Principles for the Application of Risk Analysis to the Storage and Transportation of Military Ammunition and Explosives," models and data, as implemented in nationally approved software tools. Each tool has its limitations and conditions for use, so these must be considered when using them to conduct a risk analysis.

examination by an ESO based on experience and knowledge. Quantitative values, as well as the relation between quantitative values and qualitative levels are provided in Table 2-7 of AASTP-5.

- d. There are a number of possible causes for an undesired explosive event. Some of the potential causes that should be considered include:
 - (1) Threat of fire, the most common cause of munitions-related events
 - (2) Human error (e.g., manning, workload, proper equipment, fatigue)
 - (3) Intruder potential (e.g., sabotage, theft, other subversive activity)
 - (4) Enemy action (e.g., improvised explosive device, indirect or direct fire threats)
 - (5) Environmental (e.g., lightning risk, high temperatures or humidity)
 - (6) Condition of the ammunition (e.g., deteriorated material, condition code, fatigue or chemical breakdown, damaged, lower standard of manufacture (i.e., captured))
- e. The risk level obtained will define the required risk approval level for deviating from the NATO explosives safety requirements of AASTP-1 or AASTP-5, as applicable (see Chapter 2).

Table 7-1. Risk Management Matrix

Consequence Probability	Catastrophic	Major	Minor	Negligible
Likely	High	High	Medium	Low
Occasional	High	Medium	Medium	Low
Seldom	Medium	Medium	Low	Low
Unlikely	Medium	Low	Low	Low
Very Unlikely	Low	Low	Low	Low

Source: AASTP-5, Table 2-6: Risk Index Table, page 2-44

Table 7-2. Event Probability Definitions

Level*	Qualitative Definition
Likely	Will occur frequently
Occasional	Will occur several times
Seldom	Unlikely, but can reasonably be expected to occur
Unlikely	Unlikely to occur, but possible
Very Unlikely	So unlikely it may be assumed it will never occur

Source: AASTP-5, Table 2-4: Event Probability (qualitative values), page 2-42

*The terms used here and their alternatives reflect the variety that are used by nations (semantics) in their risk management processes. Some nations' processes may vary from three to five levels.

Table 7-3. Consequence Definitions (Qualitative Values)

Category	Definition
Catastrophic	• Severe consequences unacceptable in all but the most urgent of operational requirements • Multiple deaths and/or serious injury • Significant loss or damage to mission critical materiel and infrastructure
Major	• Critical consequences and acceptance implies operational imperatives • Some deaths and/or injury • Loss or damage to mission critical materiel and infrastructure
Minor	• Consequences are not expected to significantly disrupt operations • Minor injuries • Minimal impact on materiel and infrastructure
Negligible	Negligible or insignificant effects

Source: AASTP-5, Table 2-5, Consequence Table (qualitative values), page 2-44

7.13. STEP 8. GENERATE THE ESMRM RISK ASSESSMENT REPORT

7.13.1. After completing the risk assessment and identifying the appropriate risk level, the assessment team lead will, in coordination with assessment team members, develop a report that includes the following:

- a. Executive summary (to include at a minimum overall risk by location, recommended risk decision, implementation of mitigation, and oversight).
- b. Purpose.
- c. Scope (with signatures and modifications).
- d. Methodology used to assess the strategic location and determine munitions and munitions-related process risks (e.g., QD, risk assessment tools, protective construction methodologies).
- e. Explosives safety supporting information (e.g., site plans, deviations, exposures).
- f. Identification of munitions and munitions-related processes.
- g. Infrastructure analysis based on risk to and from munitions and munitions-related processes.

Overall risks to and from munitions and munitions-related processes.

- h. A Control Plan (see paragraph 2.7.2 of AASTP-5 for details) that shall contain recommendations for mitigating munitions risks to protect people, property, and the environment, while sustaining and maximizing operational capabilities and readiness to as low as reasonably practicable.
- i. Organizations/agencies/units responsible for implementing and managing risk-reduction actions. Elements of managing a risk decision include, but are not limited to:
 - (1) Tracking the implementation of the risk-reducing actions.
 - (2) Need for reassessment of the ESMRM risk decision based on conditions specified in paragraph 7.3.
 - (3) Recommending when the ESMRM risk decision is no longer needed due to the conclusion of a munitions-related operation.
- j. Organizations/agencies/units responsible for overseeing approved risk reduction actions.
- k. Estimated duration of the ESMRM risk decision.
- l. Additional information, as required.

7.13.2. Team lead provides the final ESMRM risk assessment report to the requesting NATO commander in support of the risk decision.

7.14. STEP 9. MAKE THE ESMRM RISK DECISION (AND APPROVE RISK REDUCTION ACTIONS)

After the ESMRM Risk Assessment Report is completed and provided to the requesting NATO commander, the risk decision can be made. Key elements of the NATO commander's risk decision include:

- a. Determining if the residual risk identified in the Control Plan is acceptable and deciding if risk-reducing actions are sufficient and acceptable. If the NATO commander determines that the risk level is too high, he/she can direct further analysis and development of additional risk-reducing actions or change or reject the course of action.
- b. Approving risk-reducing actions identified in the Control Plan.
- c. Identifying/designating organizations/agencies/units responsible for implementing, tracking, maintaining, and overseeing the risk-reducing actions contained in the ESMRM report.

7.15. STEP 10. COORDINATE AND COMMUNICATE, IMPLEMENT, TRACK, AND MAINTAIN THE NATO COMMANDER'S ESMRM RISK DECISION

7.15.1. Coordination and Communication. The NATO commander's risk decision shall be coordinated and communicated with all impacted parties throughout the appropriate chain of command to communicate the hazards and controls to be implemented. Chapter 2 provides additional chain of command details.

7.15.2. Implementing the NATO Commander's Risk Decision. The NATO commander shall identify/designate organizations/agencies/units responsible for implementing the risk decision.

7.15.3. Tracking the NATO Commander's Risk Decision. The NATO commander shall identify/designate organizations/agencies/units responsible for tracking risk-reducing actions.

7.15.4. Maintenance. The organizations/agencies/units responsible for implementing and managing risk reduction actions shall determine when conditions require reevaluation as stated in paragraph 7.3, or when the risk decision is no longer required (e.g., a risk decision at a commercial seaport of debarkation when the operation is complete).

7.16. STEP 11. PROVIDE OVERSIGHT OF THE ESMRM RISK DECISION

The organizations/agencies/units responsible for overseeing the risk reduction actions shall report periodically through the chain of command to the NATO commander regarding the status and continued need for the risk decision and mitigation. The overseeing organization shall notify the decision maker, through the chain of command, when the risk decision is no longer needed.

7.17. STEP 12. TRACK LESSONS LEARNED

7.17.1. Lessons Learned and Information Management. Improvement of the risk assessment process can be achieved by capturing and leveraging lessons learned, as well as effectively managing and maintaining the data from the assessments process. Risk assessment information and the risk decisions should be provided to NATO planners for integration into plans, training exercises, and operational documents.

7.17.2. Repository. The AC/305 ESMRM Panel shall maintain a repository of ESMRM risk decisions and supporting assessments/reports.

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ANNEX A ESMRM RISK ASSESSMENT FORM

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Assessment Type and Operation					
1. <u>Type of ESMRM Risk Assessment</u> (i.e., 1. Planning; 2. RSOI; 3. Storage; 4. Transportation; 5. Distribution/Collection; 6. Maintenance and Handling; 7. Retrograde and Removal; or 8. Disposal and Demilitarization):					
2. <u>Military Munitions Operations</u> (i.e., planning, training exercise, or operation):					
3. <u>Location</u> (e.g., country, port, railhead, NATO operating base, training area):					
Potential Explosion Site (PES)					
4. <u>GPS Coordinates</u> :			5. <u>Length</u> :		
			<u>Width</u> :		
			<u>Area</u> :		
7. <u>Hazard Division (HD) and Class</u>		8. <u>Explosives Quantity</u>		6. <u>Any Additional PES Related Information</u> :	
		9. Inhabited Building Distance (IBD)			
	<u>Assessed</u>	<u>Actual</u>	<u>Assessed</u>		<u>Actual</u>
1.1					
1.2.1					
1.2.2					
1.2.3					
1.3.1					
1.3.2					
1.4					

Summary of Exposed Sites (ESs) and Population			
ES Facilities and Assets Data			
<u>Types of Facilities & Assets within IBD*</u>	<u>Number of Facilities and Assets within IBD</u>	<u>Approximate Cost</u>	<u>Comments:</u>
<u>Military</u>			
<u>Civilian</u>			
<u>Total</u>			
ES Population Data			
<u>Types of Facilities and Assets within IBD*</u>	<u>Unrelated Personnel within IBD</u>	<u>Related Personnel within IBD</u>	<u>Comments:</u>
<u>Military</u>			
<u>Civilian</u>			
<u>Total</u>			

*For example, commercial buildings, exposed personal and commercial vehicles, petroleum storage tanks, power stations.

ANNEX B ESMRM Risk Assessment Form - Example
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Assessment Type and Operation					
1. <u>Type of ESMRM Risk Assessment</u> (i.e. 1. Planning 2. RSOI 3. Storage 4. Transportation 5. Distribution/Collection 6. Maintenance and Handling 7. Retrograde and Removal or 8. Disposal and Demilitarization): Transportation					
2. <u>Military Munitions Operations</u> (i.e. planning, training exercise, or operation): Exercise Capable Logistician 2025					
3. <u>Location (e.g. country, port, railhead, NATO operating base, training area):</u> United States, Port of Valdez Alaska (Commercial)					
Potential Explosion Site (PES)					
4. <u>GPS Coordinates:</u> 123456			5. <u>Length:</u> 400m <u>Width:</u> 20m <u>Area:</u> 8,000m		
7. <u>Hazard Division (HD) and Class</u>		8. <u>Explosives Quantity</u>		6. <u>Any Additional PES Related Information:</u> Commercial port has an existing Hazard Division 1.1 operating license.	
		<u>Assessed</u>	<u>Actual</u>		<u>Assessed</u>
9. <u>Inhabited Building Distance (IBD)</u>					
1.1	20,000kg	10,000kg	400m		400m
1.2.1	-	-			
1.2.2	-	-			
1.2.3	-	-			
1.3.3	-	-			
1.3.4	-	-			
1.4	-	-			

Summary of Exposed Sites (ES) and Population			
ES Facilities and Assets Data			
<u>Types of Facilities & Assets within IBD</u> <u>(e.g. commercial buildings, exposed personal and commercial vehicles, petroleum storage tanks, power stations)</u>	<u>Number of Facilities and Assets within IBD</u>	<u>Approximate Cost</u>	<u>Comments:</u> Port of Valdez is a commercial port therefore there is no NATO base involved. 15 assets consist of 10 commercial buildings -(7 Town of Valdez, 3 commercial fishing processing buildings) 1 pier 1 grain storage silo 1 private residence 1 port refueling facility 1 Public Traffic Route (State Road)
<u>Military</u>	N/A	N/A	
<u>Civilian</u>	15	2,000,000 €	
<u>Total</u>	15	2,000,000 €	
ES Population Data			
	<u>Unrelated Personnel within IBD</u>	<u>Related Personnel within IBD</u>	<u>Comments:</u> Recommend that as part of the risk mitigating solutions that unrelated personnel vacate commercial and private locations within IBD during munitions operations at the port.
<u>Military</u>	N/A	N/A	
<u>Civilian</u>	200	50	
<u>Total</u>	200	50	Discussions with City and Port Authority personnel indicate that they are willing to accommodate the recommended risk mitigating recommendation.

ANNEX C REFERENCES

Allied Administrative Publications (AAPs)

AAP-06, Edition 2013, Version 2, "NATO Glossary of Terms and Definitions (English & French)"

AAP-15, Edition 2013, Version 1, "NATO Glossary of Abbreviations Used in NATO Documents and Publications"

Allied Ammunition Storage and Transport Publications (AASTPs)

AASTP-1, Edition 1, Change 3, "Manual of Safety Principles for the Storage of Military Ammunition and Explosives"

AASTP-3, Edition 1, Change 3, "Manual of NATO Safety Principles for the Hazard Classification of Military Ammunition and Explosives"

AASTP-4, Edition 1, Version 3, "Manual on the NATO Principles for the Application of Risk Analysis to the Storage and Transportation of Military Ammunition and Explosives"

AASTP-5, Edition 1, Version 2, "NATO Guidelines for the Storage, Maintenance and Transport of Ammunition on Deployed Missions or Operations"

Allied Command Operations (ACO) "Comprehensive Operations Planning Directive" (COPD)

Allied Committee (AC)/305 (EAPC) D(2013)0008-AS11 29 April 2013, "NATO Explosives Safety and Munitions Risk Management (ESMRM) Policy"

Allied Joint Publications (AJPs)

AJP-01, "Allied Joint Doctrine"

AJP-3, "Allied Joint Doctrine for the Conduct of Operations"

AJP-3.2, "Allied Joint Doctrine for Land Operations"

AJP-3.4, "Allied Joint Doctrine for Non-Article 5 Crisis Response Operations"

AJP-3.4.1, "Peace Support Operations"

AJP-3.5, Edition A, Version 1, "Allied Joint Doctrine for Special Operations"

AJP-3.12, "Allied Doctrine for Military Engineer Support to Joint Operations"

AJP-3.13, "Allied Joint Doctrine for the Deployment of Forces"

AJP-3.14, "Allied Joint Doctrine for Force Protection"

AJP-4, "Allied Joint Logistic Doctrine"

AJP-4.4, "Allied Joint Movement and Transportation Doctrine"

AJP-4.5, "Allied Joint Doctrine for Host Nation Support"

AJP-4.9, "Allied Joint Doctrine for Modes of Multinational Logistic Support"

AJP-4.10, "Allied Joint Medical Support Doctrine"

AJP-5, "Allied Joint Doctrine for Operational-Level Planning"

Allied Logistic Publication (ALP)-4.2, "Land Forces Logistic Doctrine"

Allied Movement Publication (AMovP)-6, "Allied Multimodal Transportation of Dangerous Goods Directive"

Allied Ordnance Publications (AOPs)

AOP-2, "The Identification of Ammunition"
AOP-38, "Glossary of Terms and Definitions Concerning the Safety and Suitability for Service of Munitions, Explosives and Related Products"
AOP-62, "In-Service Surveillance of Munitions General Guidance"
AOP-63, "In-Service Surveillance of Munitions Sampling and Test Procedures"
AOP-64, "In-Service Surveillance of Munitions Condition Monitoring of Energetic Materials"

Council Memorandum (C-M)(2002)60, "The Management of Non-Classified NATO Information"

Standardization Agreements (STANAGs)

STANAG 2389, "Minimum Standards of Proficiency for Trained Explosive Ordnance Disposal Personnel" (draft)
STANAG 2617, "Explosives Safety and Munitions Risk Management (ESMRM) in NATO Planning, Training, and Operations"
STANAG 2953, "The Identification of Ammunition"
STANAG 4441, "Allied Multi-Modal Transportation Of Dangerous Goods Directive"
STANAG 4518, "Safe Disposal of Munitions, Design Principles and Requirements, and Safety Assessment."
STANAG 4675, "In-Service Surveillance (ISS) of Non Nuclear Munitions"

U.N. Recommendations on the Transport of Dangerous Goods, Model Regulations, Volumes I and II, 18th Revised Edition, 2013

ANNEX D ABBREVIATIONS AND ACRONYMS

This annex lists abbreviations and acronyms used in this document as well as others commonly used in joint and combined operations. A comprehensive list is provided in AAP-15, "NATO Glossary of Abbreviations Used in NATO Documents and Publications."

AAP	Allied administrative publication
AASTP	Allied ammunition storage and transport publication
ACO	Allied Command Operations
AJP	Allied joint publication
ALARA	as low as reasonably achievable
ALARP	as low as reasonably practicable
ALP	Allied logistic publication
AMovP	Allied movement publication
AOI	area of interest
AOO	area of operations
AOP	Allied ordnance publication
APOD	airport of debarkation
APOE	airport of embarkation
AT	ammunition technician
ATO	ammunition technical officer
ATP	Allied tactical publication
BLAHA	basic load ammunition holding area
BLSA	basic load storage area
BOSI	base operations support integrator
CALA	combat aircraft loading area
CAPA	combat aircraft parking area

CEA	captured enemy ammunition
COA	course of action
CONOPLAN	contingency operation plan
COPD	Comprehensive Operations Planning Directive
CSS	combat service support
DU	depleted uranium
EAPC	Euro-Atlantic Partnership Council
EOD	explosive ordnance disposal
ERW	explosive remnants of war
ESB	Explosives Safety Board
ESMRM	Explosives Safety and Munitions Risk Management
ESO	explosives safety officer
FARP	forward arming and refueling point
FD	field distance
FLPC	Final Logistics Planning Conference
FLB	forward logistic base
FOB	forward operating base
HD	hazard division
HNS	host-nation support
HQ	headquarters
IBD	inhabited building distance
ID	initiating directive
ILPC	Initial Logistics Planning Conference
IMD	intermagazine distance

ISO	International Organization for Standardization
JALLC	Joint Analysis Lessons Learned Centre
JFC	joint force command
JTF	joint task force
LC	Logistics Committee
LOC	lines of communications
LPT	logistics preparation of the theatre
M&T	movement & transport
MLPC	Main Logistics Planning Conference
MN	multinational
MOA	memorandum of agreement
NA5CRO	non-Article 5 crisis response operation
NAC	North Atlantic Council
NATO	North Atlantic Treaty Organization
NEQ	net explosive quantity
NSA	NATO Standardization Organization
OB	open burning
OD	open detonation
OLRC	Operations and Logistics Review Conference
OPLAN	operation plan
OPP	operational planning process
OTAN	Organisation du traité de l'Atlantique Nord
QD	quantity-distance
RSOI	reception, staging, onward movement and integration

SC	strategic command
SHAPE	Supreme Headquarters Allied Powers Europe
SME	subject matter expert
SOP	standard operating procedures
SPOD	seaport of debarkation
SsD	storage sub-division
STANAG	standardization agreement
TOR	terms of reference
UN	United Nations

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